

WHAM-XREX-PFMG474 Channel/Demand Stress Matrix (Closed Loop)

Generated by `python/run_simulator_validation.py`

September 21, 2026

Purpose

All 15 non-empty channel-enable combinations x 3 demand levels (600, 3000, 5400 A – 10/50/90% of PFM_MAX_CURRENT_A_PER_CHANNEL), CLOSED loop ($K_p=0.5$, $K_i=750$, $K_d=0$ – confirmed via a live single-channel stability check the same day: clean convergence, no overshoot, essentially exact flat-top tracking), each shot a real 10s ramp / 5s flat-top / 10s ramp-down profile (25s total), 45 shots total. Pass criterion: every channel's own flat-top MEDIAN $|measured - setpoint|$ (robust to the rare single-sample PID:LOGDATA? transmission glitch documented elsewhere in this project – NOT a raw max, which a lone glitch would fail unfairly) stays under 1500 Hz, measured over the flat-top window with 0.5s trimmed off each end (the filter's own settling tail). Full raw data for every shot is archived under shots/ in this repository (gitignored).

Summary

45/45 shots PASSED.

Tabulated Results

Channels	Demand (A)	Worst channel median error (Hz)	Verdict
1	600	468	PASS
1	3000	369	PASS
1	5400	409	PASS
2	600	858	PASS
2	3000	385	PASS
2	5400	363	PASS
3	600	831	PASS
3	3000	344	PASS
3	5400	458	PASS
4	600	848	PASS
4	3000	377	PASS
4	5400	409	PASS
1+2	600	886	PASS
1+2	3000	417	PASS

Channels	Demand (A)	Worst channel median error (Hz)	Verdict
1+2	5400	409	PASS
1+3	600	874	PASS
1+3	3000	417	PASS
1+3	5400	409	PASS
1+4	600	700	PASS
1+4	3000	385	PASS
1+4	5400	360	PASS
2+3	600	877	PASS
2+3	3000	393	PASS
2+3	5400	410	PASS
2+4	600	854	PASS
2+4	3000	393	PASS
2+4	5400	409	PASS
3+4	600	875	PASS
3+4	3000	401	PASS
3+4	5400	409	PASS
1+2+3	600	909	PASS
1+2+3	3000	401	PASS
1+2+3	5400	363	PASS
1+2+4	600	906	PASS
1+2+4	3000	385	PASS
1+2+4	5400	409	PASS
1+3+4	600	913	PASS
1+3+4	3000	377	PASS
1+3+4	5400	363	PASS
2+3+4	600	851	PASS
2+3+4	3000	385	PASS
2+3+4	5400	410	PASS
1+2+3+4	600	890	PASS
1+2+3+4	3000	401	PASS
1+2+3+4	5400	409	PASS

Per-Shot Detail

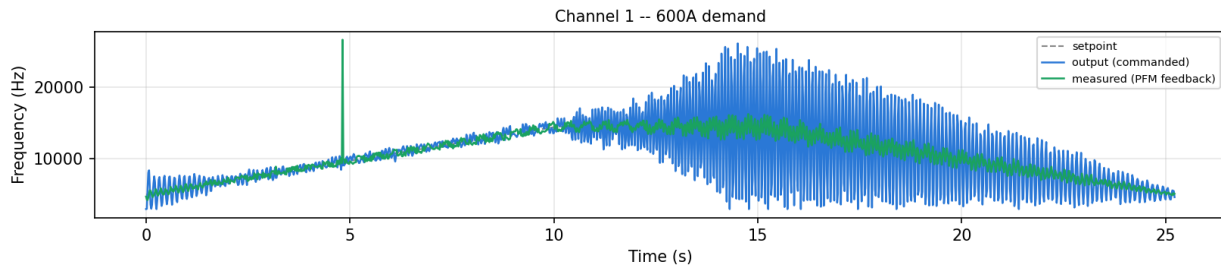
1 Shot 1: Channels 1, 600 A demand

Gated channel(s) 1 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 468 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured-setpoint|$ per channel: ch1=468Hz.

Verdict: PASS

1.1 Per-channel response

Matrix: channels 1, 600 A demand -- PASS (worst median error 468 Hz)

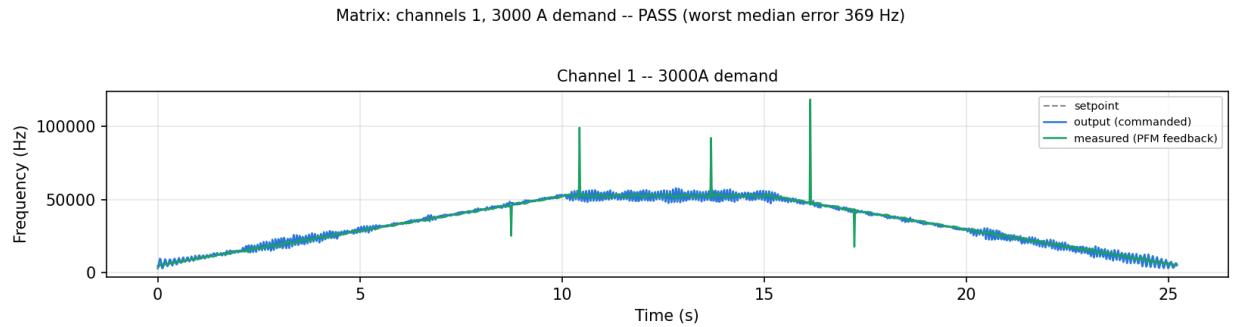


2 Shot 2: Channels 1, 3000 A demand

Gated channel(s) 1 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 369 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured-setpoint|$ per channel: ch1=369Hz.

Verdict: PASS

2.1 Per-channel response

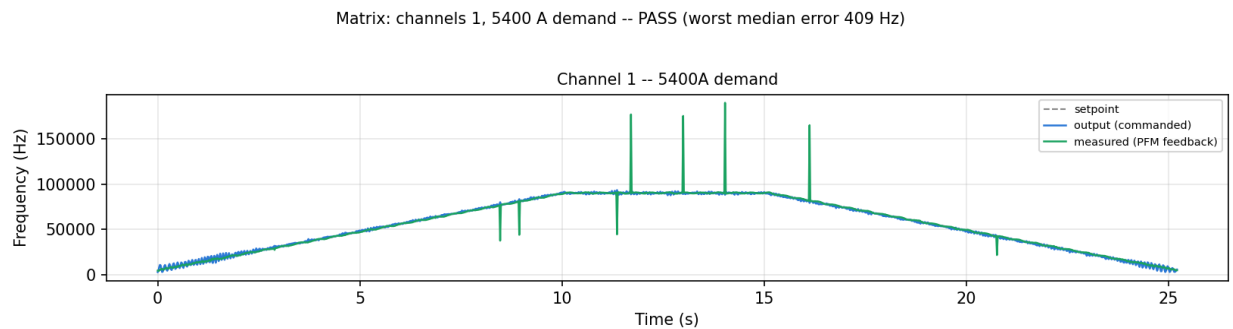


3 Shot 3: Channels 1, 5400 A demand

Gated channel(s) 1 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=409Hz.

Verdict: PASS

3.1 Per-channel response

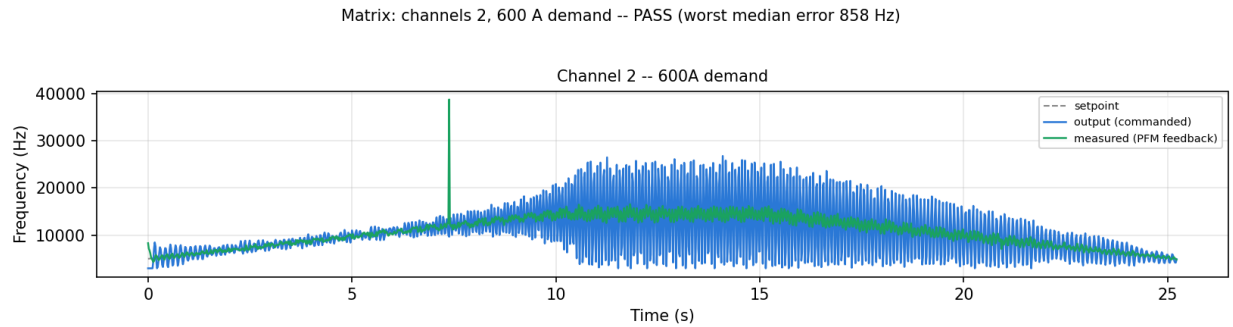


4 Shot 4: Channels 2, 600 A demand

Gated channel(s) 2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 858 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=858Hz.

Verdict: PASS

4.1 Per-channel response

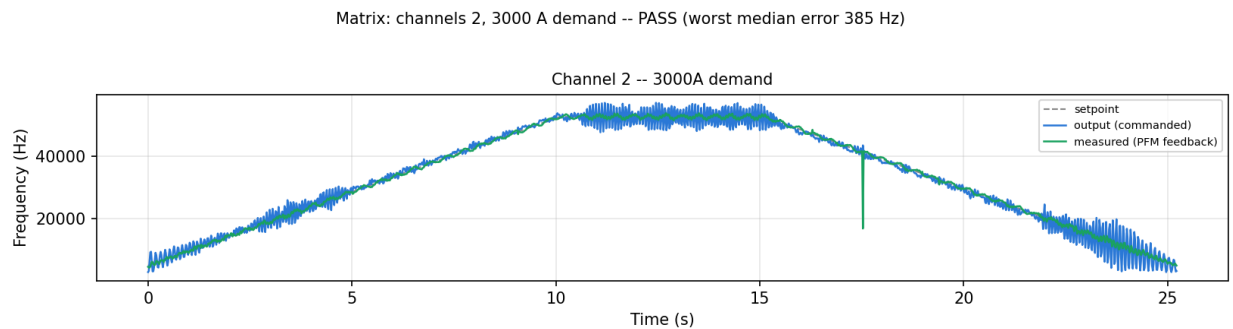


5 Shot 5: Channels 2, 3000 A demand

Gated channel(s) 2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 385 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=385Hz.

Verdict: PASS

5.1 Per-channel response



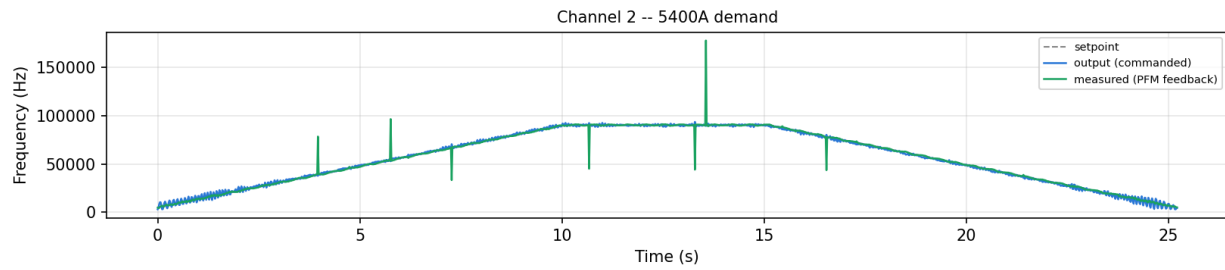
6 Shot 6: Channels 2, 5400 A demand

Gated channel(s) 2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 363 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=363Hz.

Verdict: PASS

6.1 Per-channel response

Matrix: channels 2, 5400 A demand -- PASS (worst median error 363 Hz)



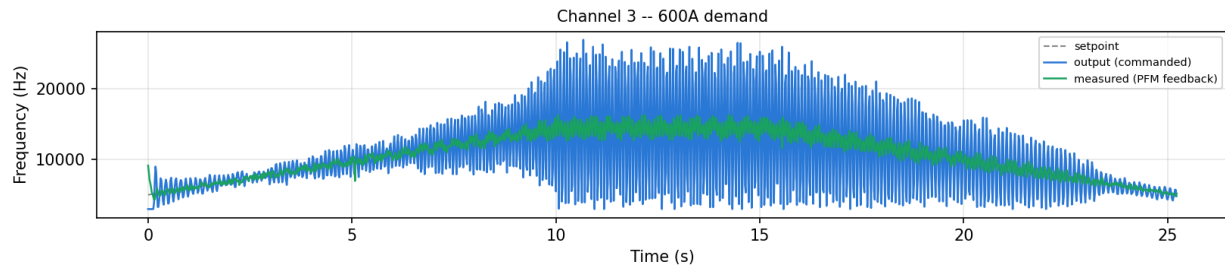
7 Shot 7: Channels 3, 600 A demand

Gated channel(s) 3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 831 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=831Hz.

Verdict: PASS

7.1 Per-channel response

Matrix: channels 3, 600 A demand -- PASS (worst median error 831 Hz)

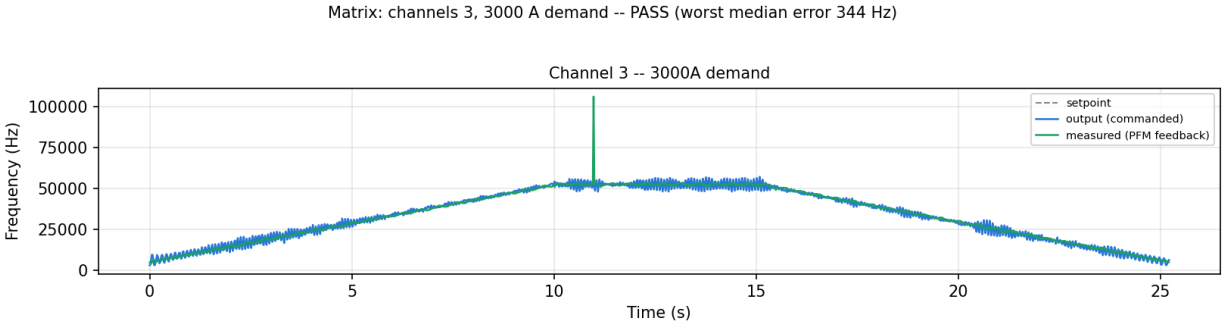


8 Shot 8: Channels 3, 3000 A demand

Gated channel(s) 3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 344 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=344Hz.

Verdict: PASS

8.1 Per-channel response

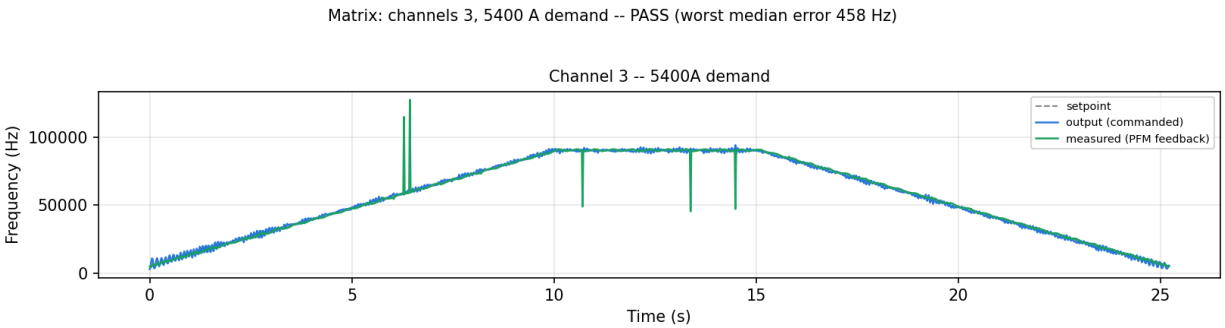


9 Shot 9: Channels 3, 5400 A demand

Gated channel(s) 3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 458 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=458Hz.

Verdict: PASS

9.1 Per-channel response

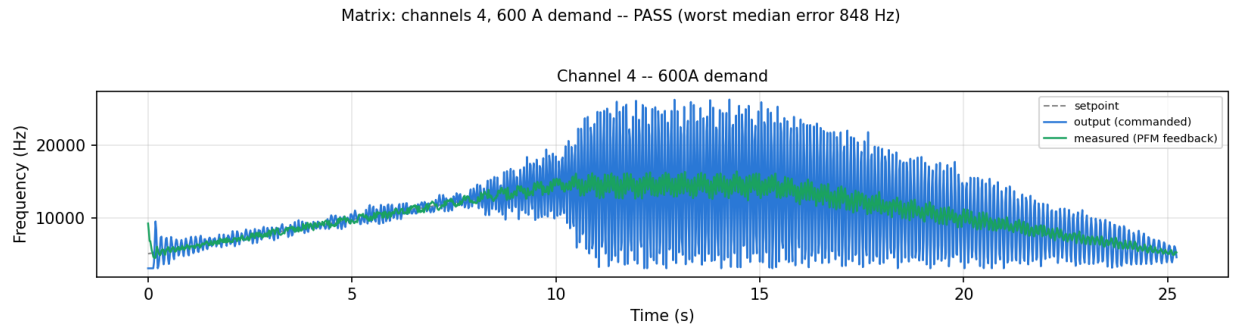


10 Shot 10: Channels 4, 600 A demand

Gated channel(s) 4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 848 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch4=848Hz.

Verdict: PASS

10.1 Per-channel response

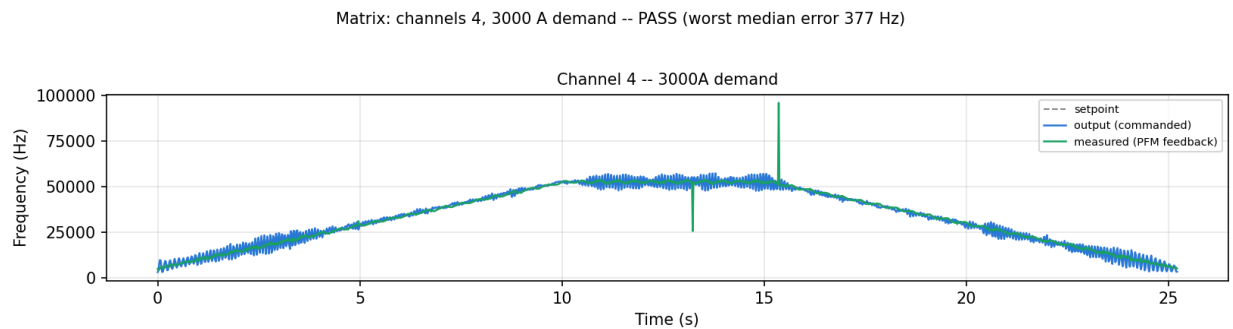


11 Shot 11: Channels 4, 3000 A demand

Gated channel(s) 4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 377 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch4=377Hz.

Verdict: PASS

11.1 Per-channel response



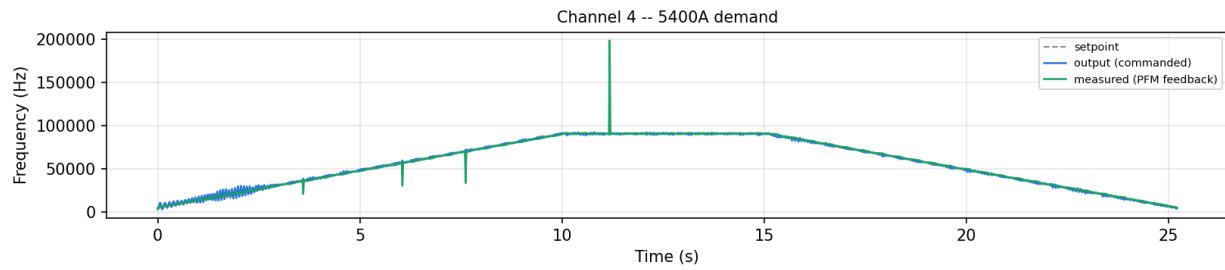
12 Shot 12: Channels 4, 5400 A demand

Gated channel(s) 4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 1 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch4=409Hz.

Verdict: PASS

12.1 Per-channel response

Matrix: channels 4, 5400 A demand -- PASS (worst median error 409 Hz)



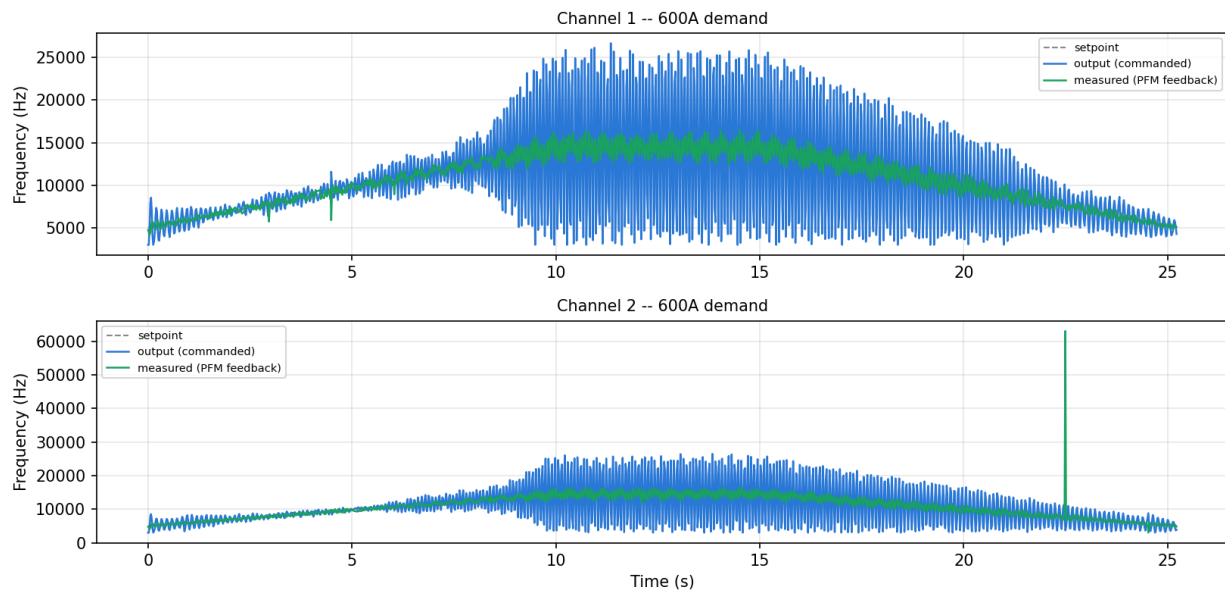
13 Shot 13: Channels 1+2, 600 A demand

Gated channel(s) 1+2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 886 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=877Hz, ch2=886Hz.

Verdict: PASS

13.1 Per-channel response

Matrix: channels 1+2, 600 A demand -- PASS (worst median error 886 Hz)



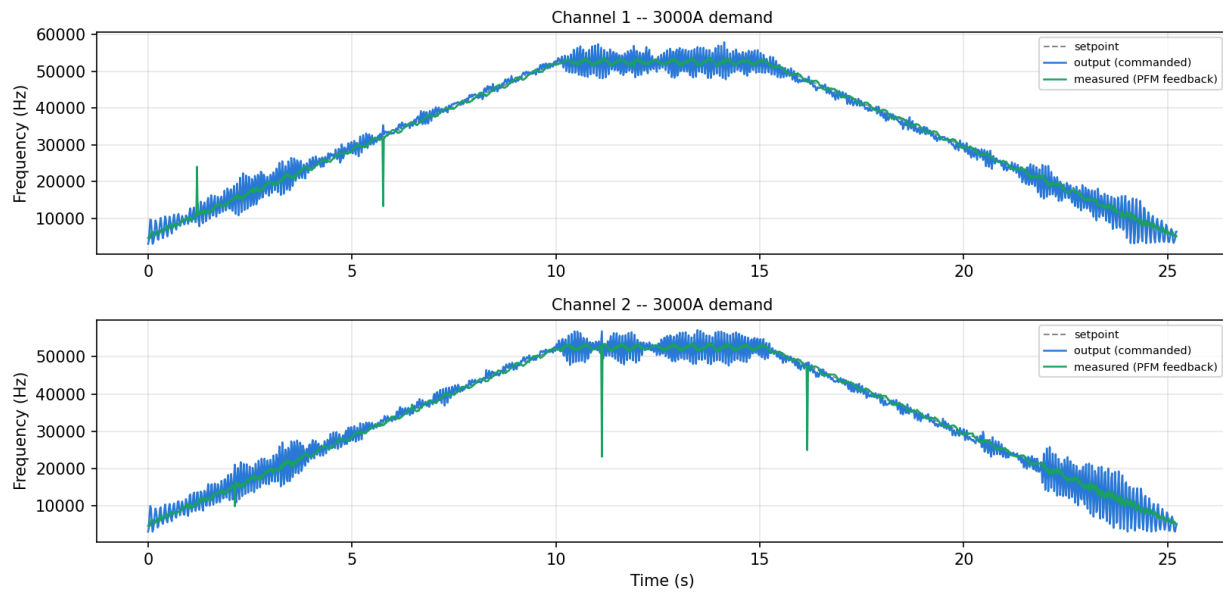
14 Shot 14: Channels 1+2, 3000 A demand

Gated channel(s) 1+2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 417 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured-setpoint|$ per channel: ch1=410Hz, ch2=417Hz.

Verdict: PASS

14.1 Per-channel response

Matrix: channels 1+2, 3000 A demand -- PASS (worst median error 417 Hz)



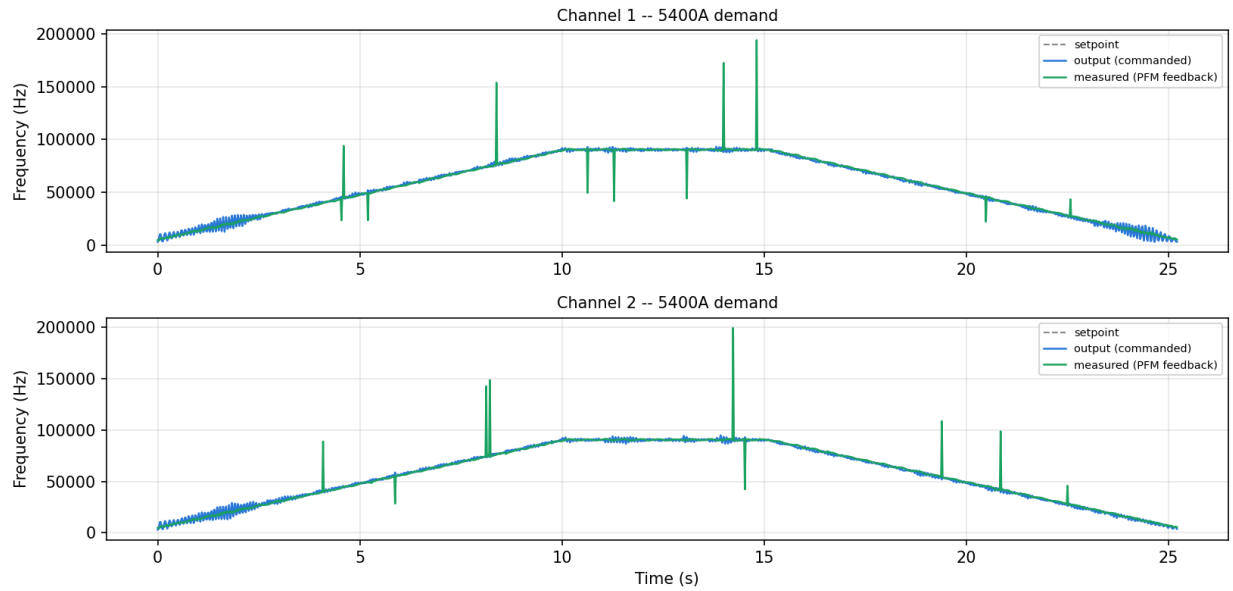
15 Shot 15: Channels 1+2, 5400 A demand

Gated channel(s) 1+2 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured-setpoint|$ per channel: ch1=409Hz, ch2=363Hz.

Verdict: PASS

15.1 Per-channel response

Matrix: channels 1+2, 5400 A demand -- PASS (worst median error 409 Hz)



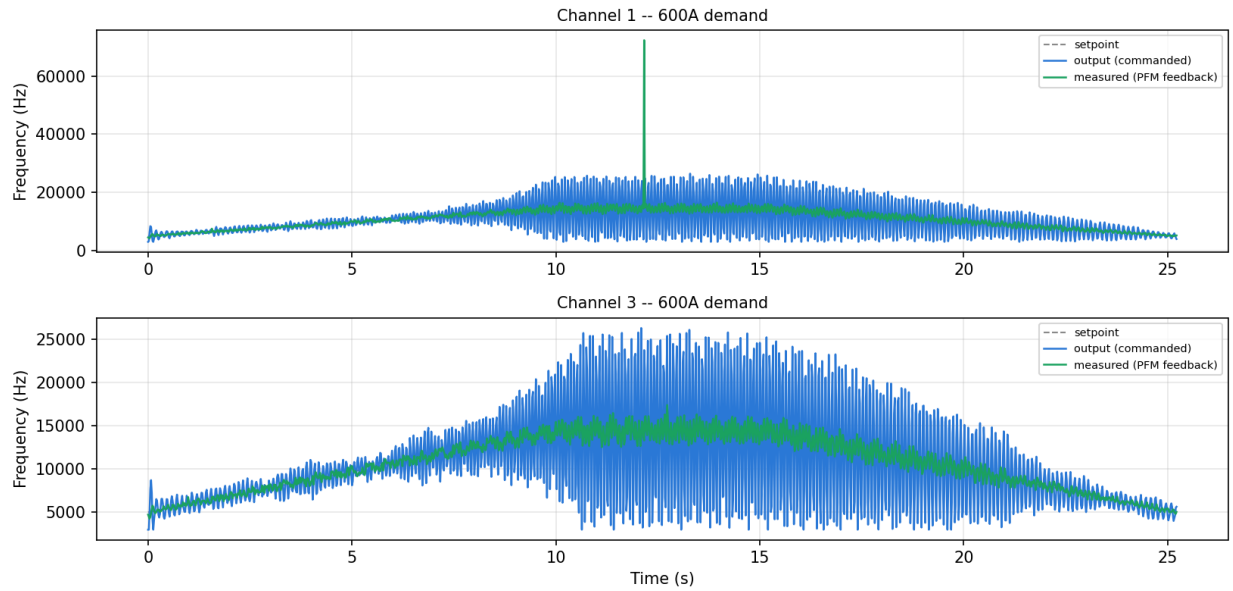
16 Shot 16: Channels 1+3, 600 A demand

Gated channel(s) 1+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 874 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=859Hz, ch3=874Hz.

Verdict: PASS

16.1 Per-channel response

Matrix: channels 1+3, 600 A demand -- PASS (worst median error 874 Hz)



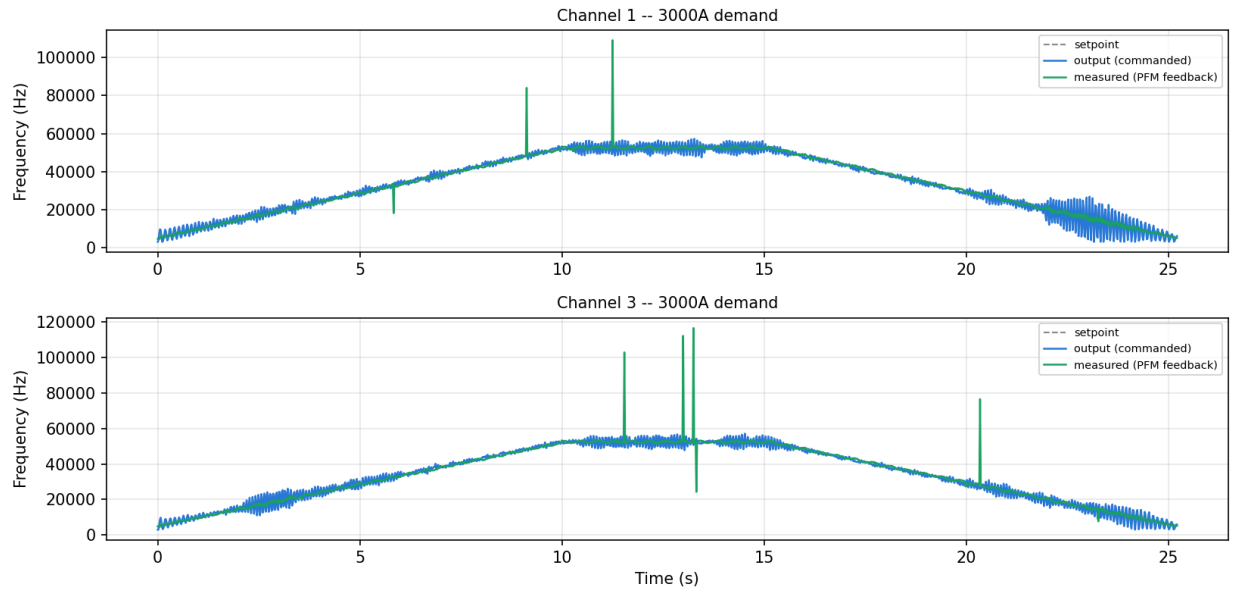
17 Shot 17: Channels 1+3, 3000 A demand

Gated channel(s) 1+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 417 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=385Hz, ch3=417Hz.

Verdict: PASS

17.1 Per-channel response

Matrix: channels 1+3, 3000 A demand -- PASS (worst median error 417 Hz)



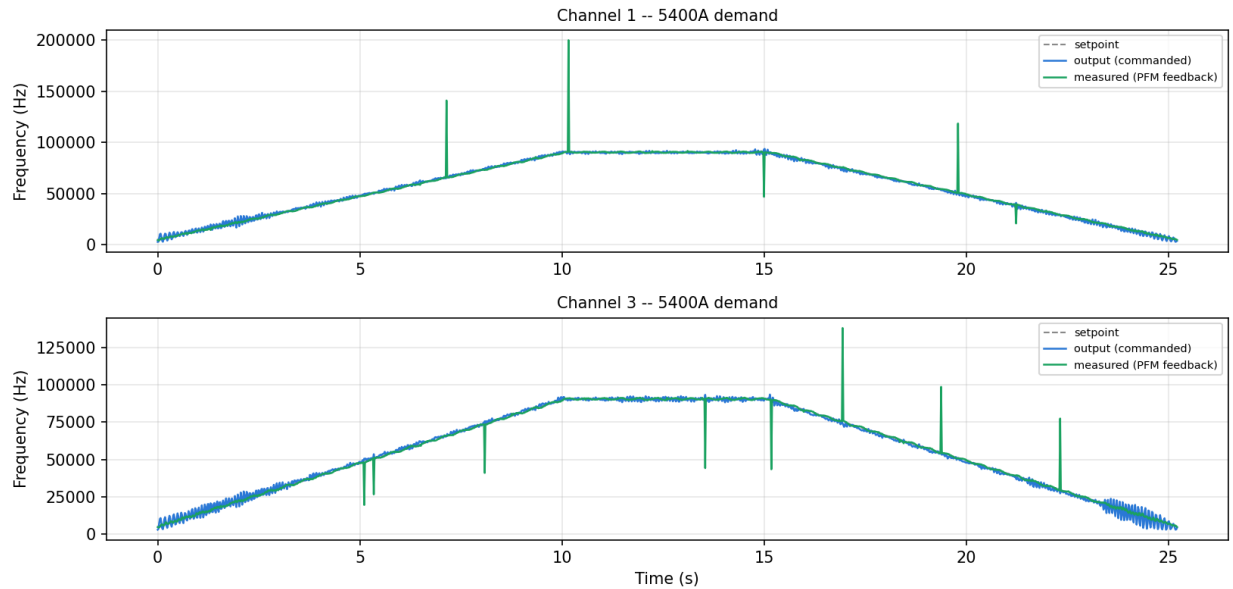
18 Shot 18: Channels 1+3, 5400 A demand

Gated channel(s) 1+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=409Hz, ch3=363Hz.

Verdict: PASS

18.1 Per-channel response

Matrix: channels 1+3, 5400 A demand -- PASS (worst median error 409 Hz)



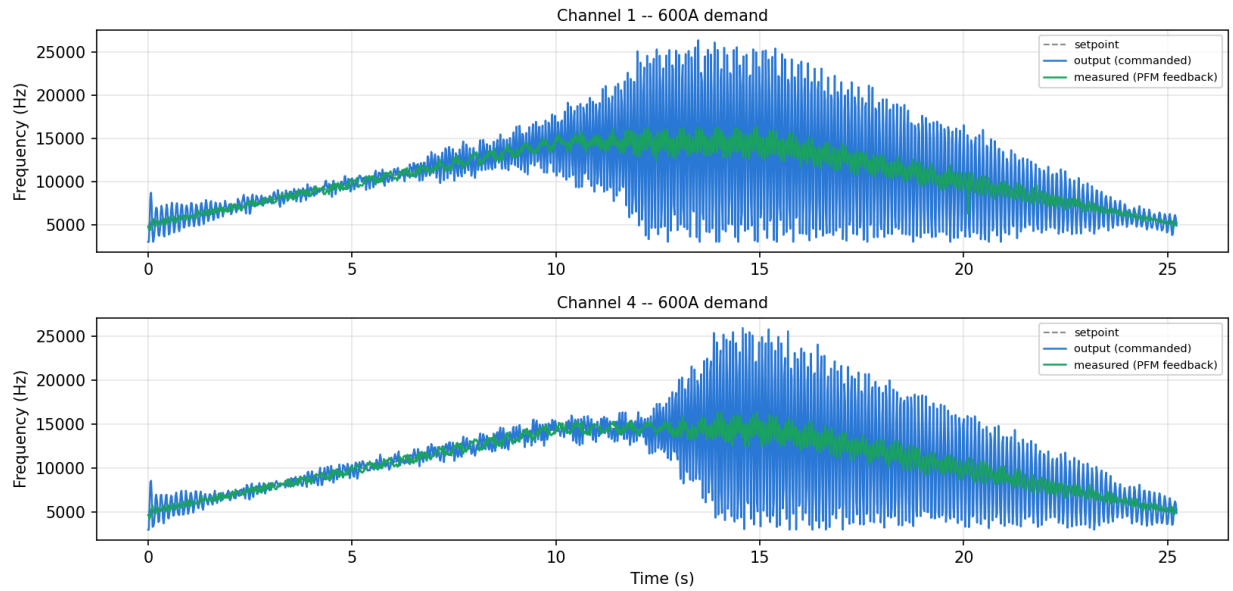
19 Shot 19: Channels 1+4, 600 A demand

Gated channel(s) 1+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 700 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=700Hz, ch4=404Hz.

Verdict: PASS

19.1 Per-channel response

Matrix: channels 1+4, 600 A demand -- PASS (worst median error 700 Hz)



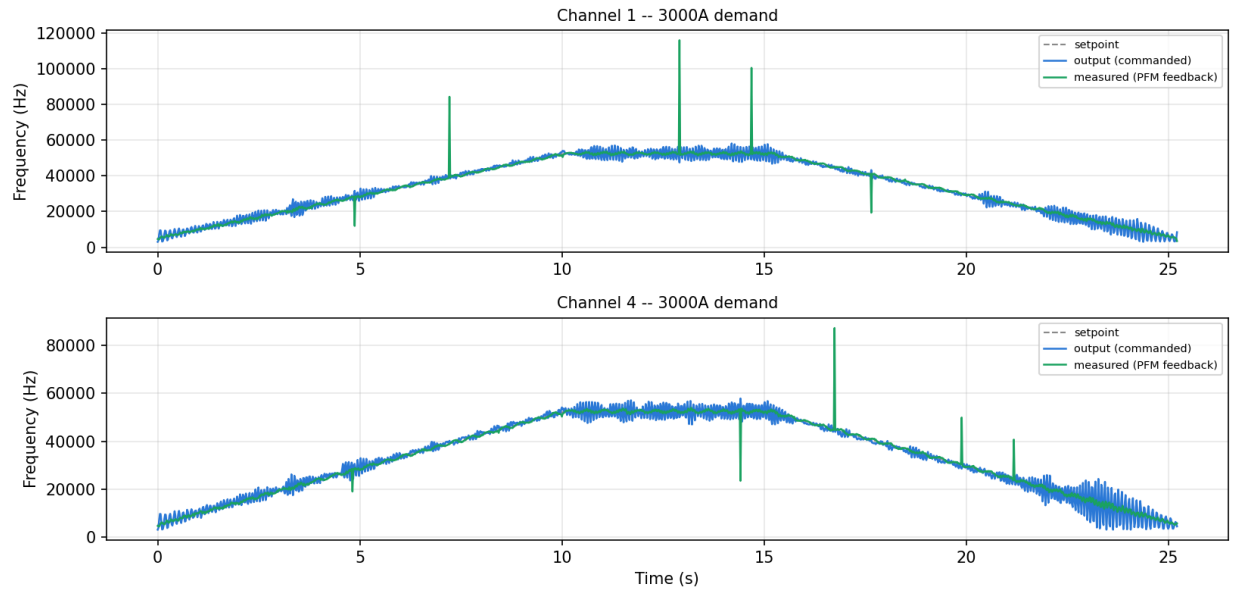
20 Shot 20: Channels 1+4, 3000 A demand

Gated channel(s) 1+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 385 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=377Hz, ch4=385Hz.

Verdict: PASS

20.1 Per-channel response

Matrix: channels 1+4, 3000 A demand -- PASS (worst median error 385 Hz)



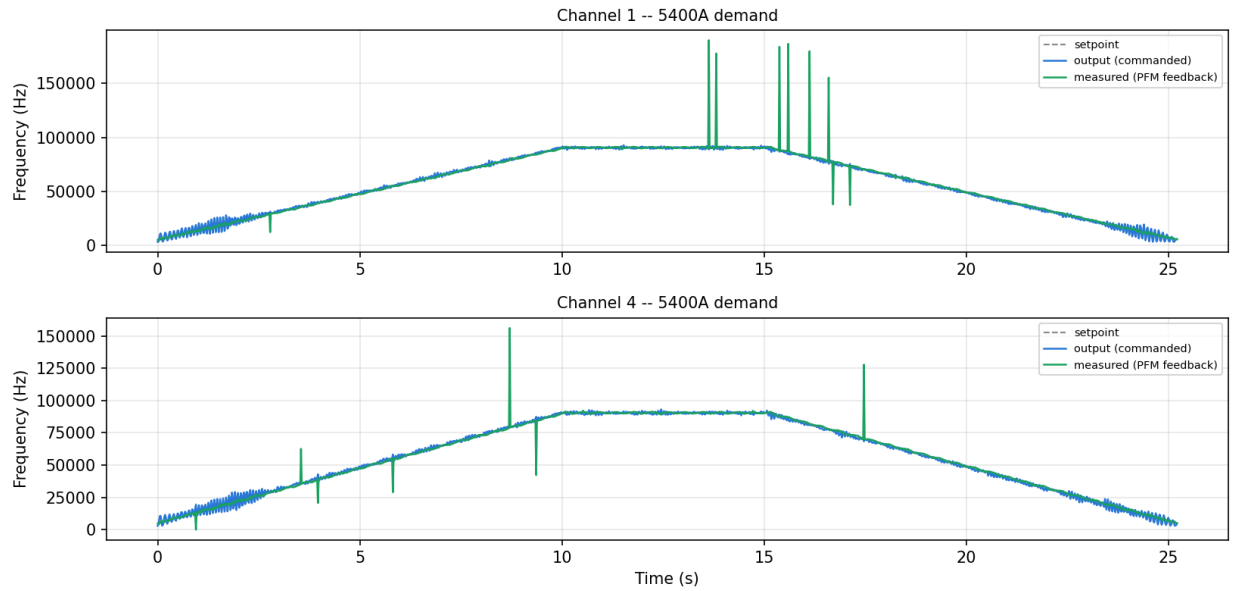
21 Shot 21: Channels 1+4, 5400 A demand

Gated channel(s) 1+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 360 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=360Hz, ch4=360Hz.

Verdict: PASS

21.1 Per-channel response

Matrix: channels 1+4, 5400 A demand -- PASS (worst median error 360 Hz)



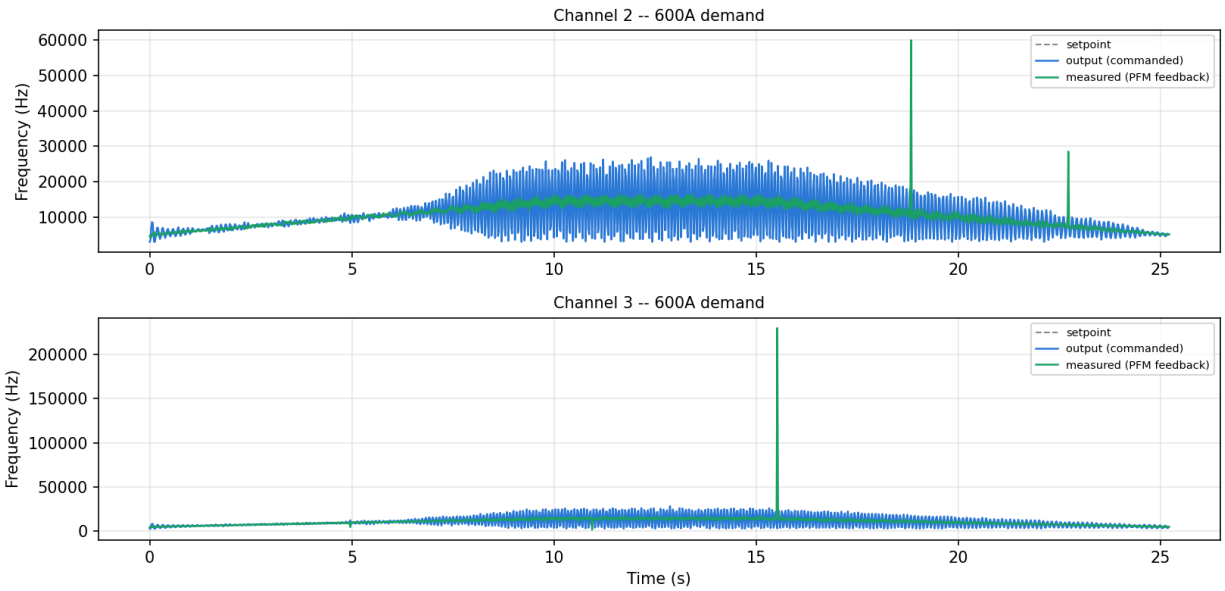
22 Shot 22: Channels 2+3, 600 A demand

Gated channel(s) 2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 877 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=877Hz, ch3=870Hz.

Verdict: PASS

22.1 Per-channel response

Matrix: channels 2+3, 600 A demand -- PASS (worst median error 877 Hz)



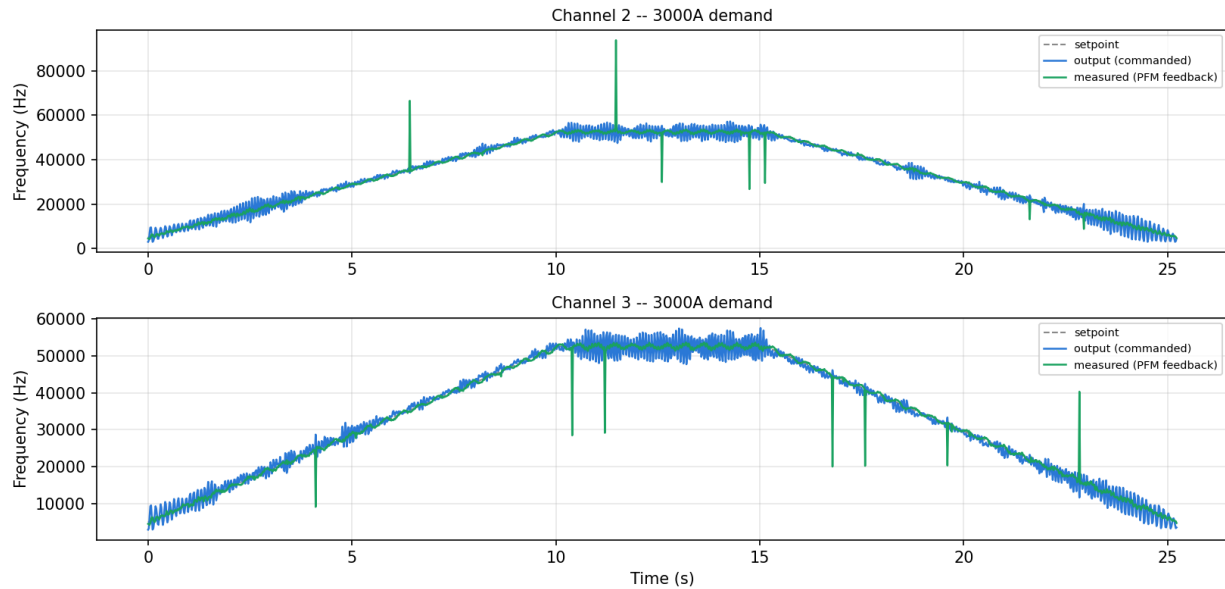
23 Shot 23: Channels 2+3, 3000 A demand

Gated channel(s) 2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 393 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=353Hz, ch3=393Hz.

Verdict: PASS

23.1 Per-channel response

Matrix: channels 2+3, 3000 A demand -- PASS (worst median error 393 Hz)



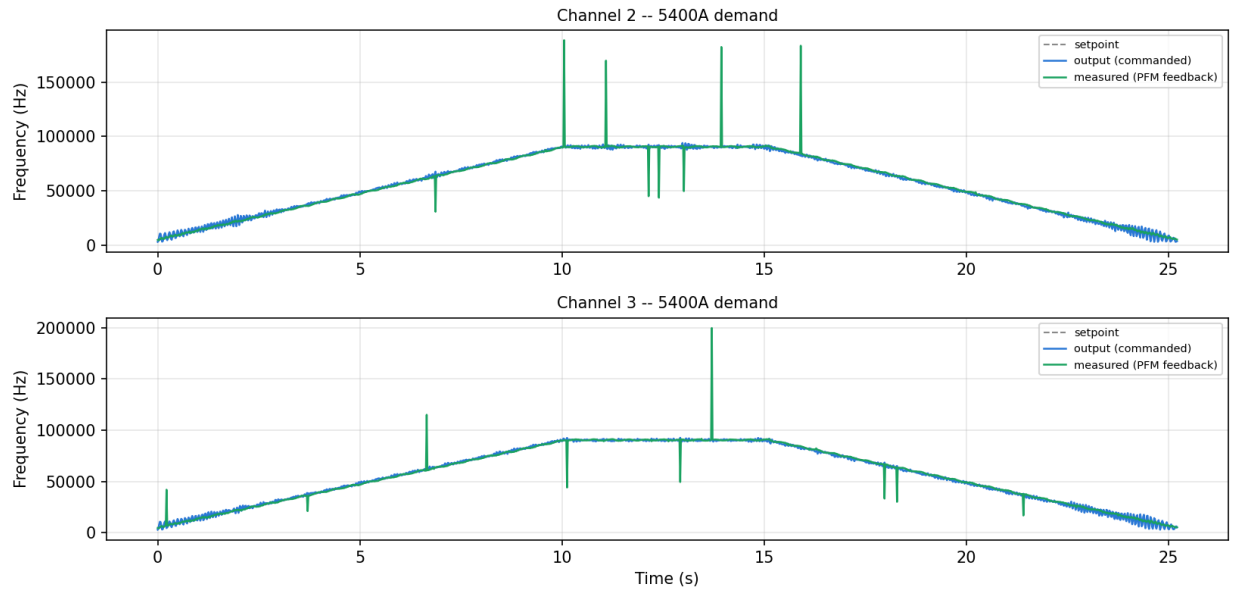
24 Shot 24: Channels 2+3, 5400 A demand

Gated channel(s) 2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 410 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=410Hz, ch3=363Hz.

Verdict: PASS

24.1 Per-channel response

Matrix: channels 2+3, 5400 A demand -- PASS (worst median error 410 Hz)



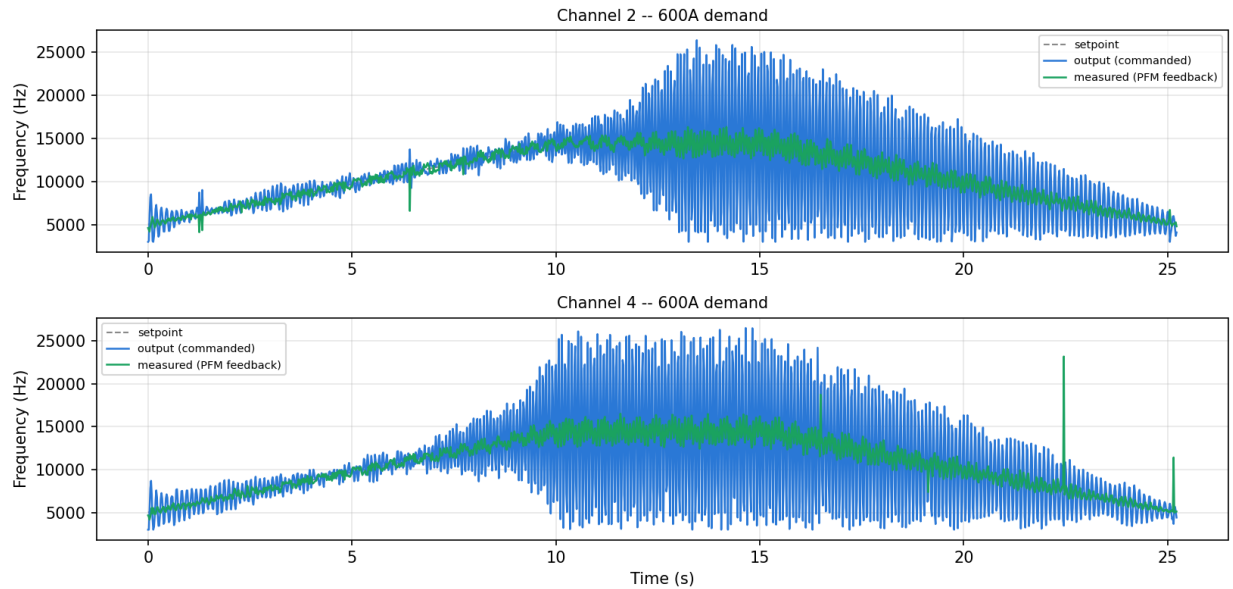
25 Shot 25: Channels 2+4, 600 A demand

Gated channel(s) 2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 854 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=533Hz, ch4=854Hz.

Verdict: PASS

25.1 Per-channel response

Matrix: channels 2+4, 600 A demand -- PASS (worst median error 854 Hz)



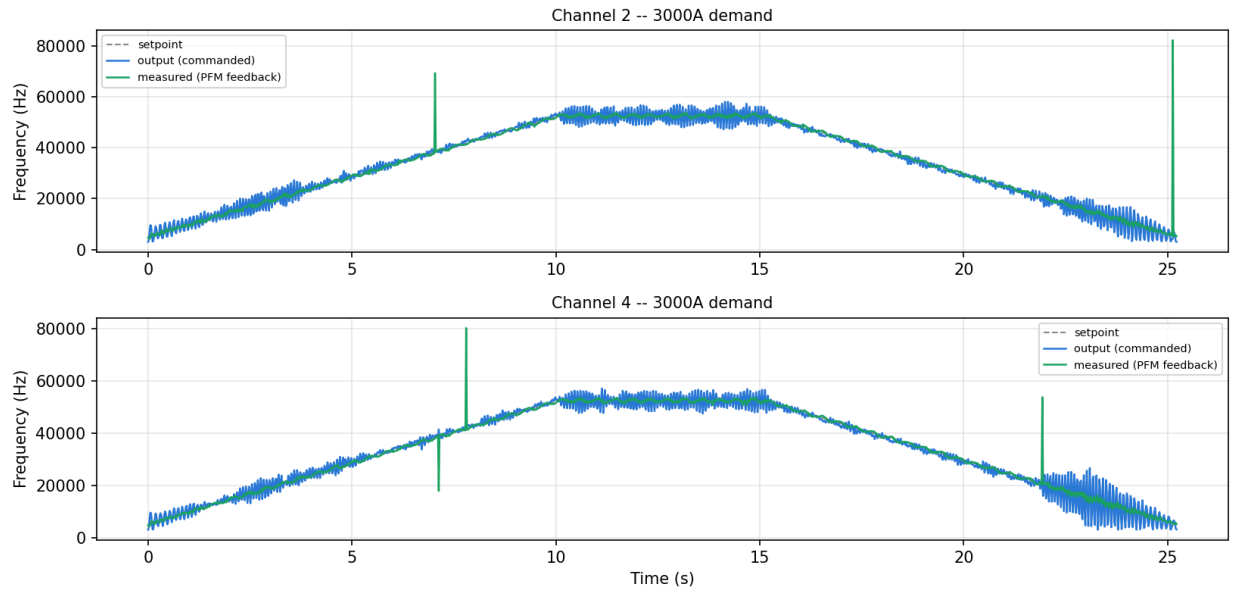
26 Shot 26: Channels 2+4, 3000 A demand

Gated channel(s) 2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 393 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=385Hz, ch4=393Hz.

Verdict: PASS

26.1 Per-channel response

Matrix: channels 2+4, 3000 A demand -- PASS (worst median error 393 Hz)



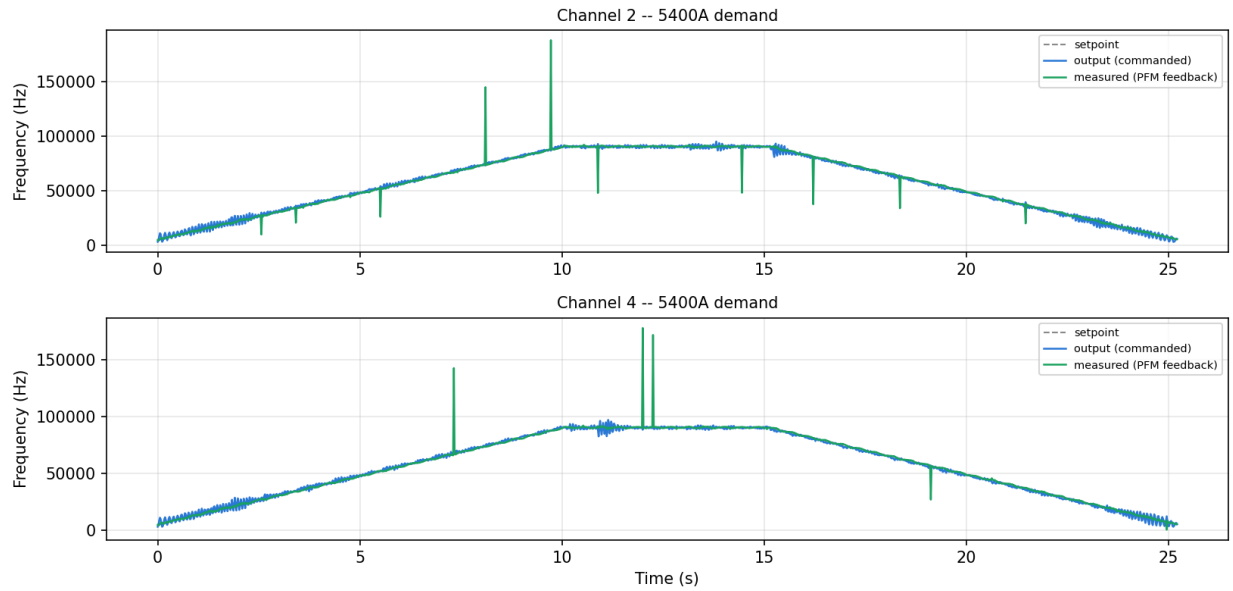
27 Shot 27: Channels 2+4, 5400 A demand

Gated channel(s) 2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=409Hz, ch4=409Hz.

Verdict: PASS

27.1 Per-channel response

Matrix: channels 2+4, 5400 A demand -- PASS (worst median error 409 Hz)



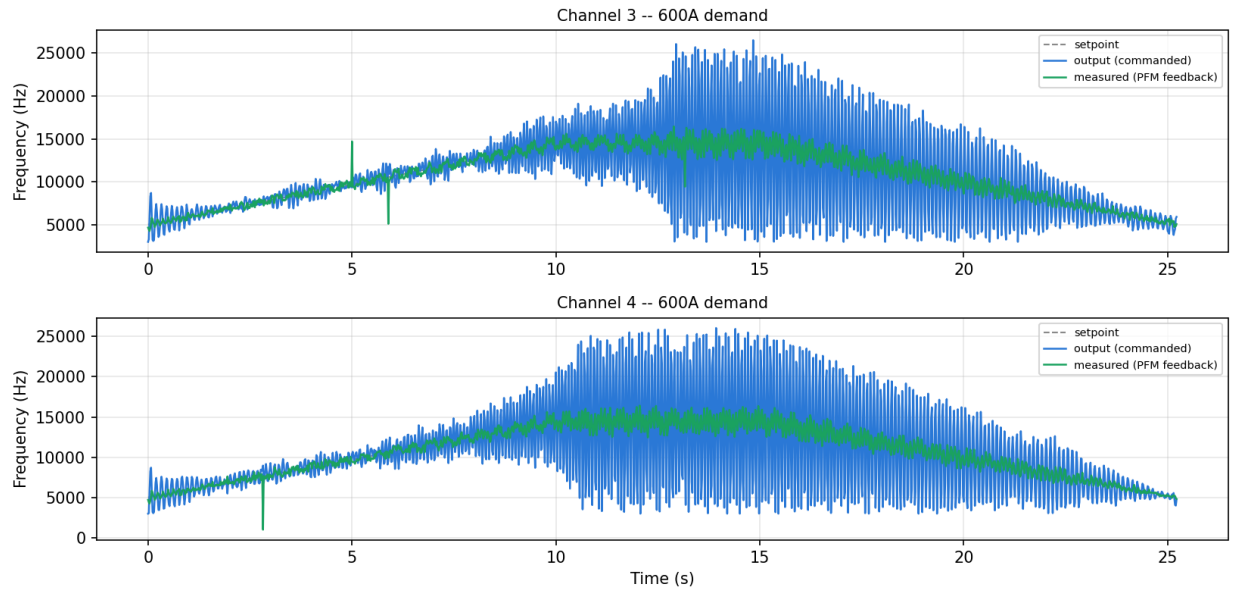
28 Shot 28: Channels 3+4, 600 A demand

Gated channel(s) 3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 875 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=572Hz, ch4=875Hz.

Verdict: PASS

28.1 Per-channel response

Matrix: channels 3+4, 600 A demand -- PASS (worst median error 875 Hz)



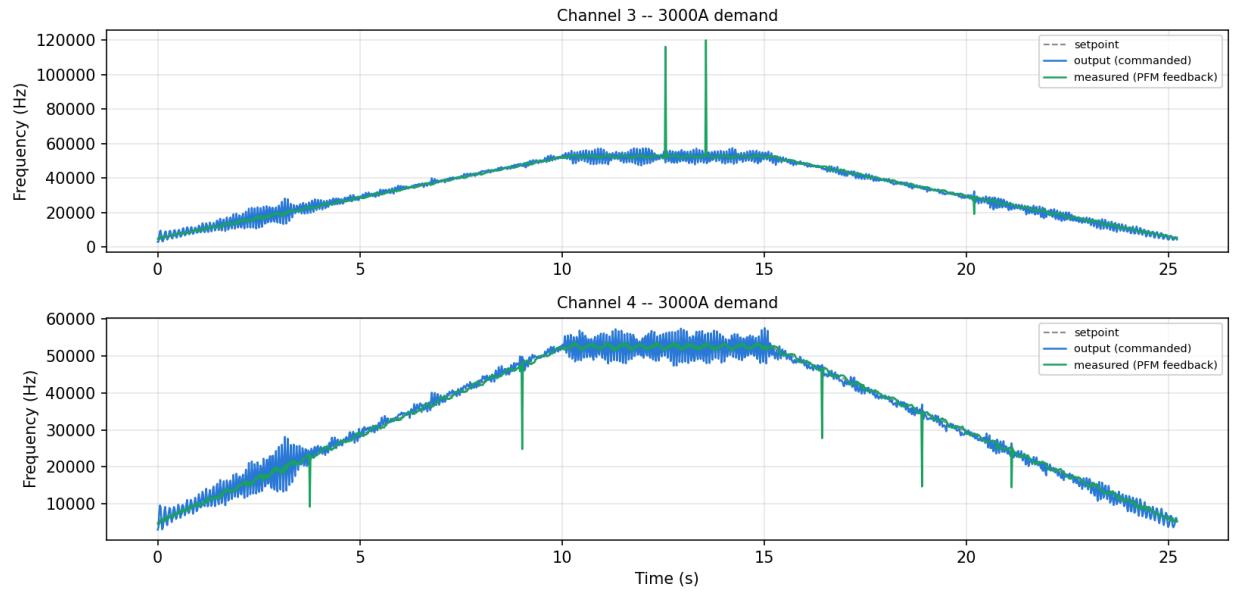
29 Shot 29: Channels 3+4, 3000 A demand

Gated channel(s) 3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 401 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=393Hz, ch4=401Hz.

Verdict: PASS

29.1 Per-channel response

Matrix: channels 3+4, 3000 A demand -- PASS (worst median error 401 Hz)



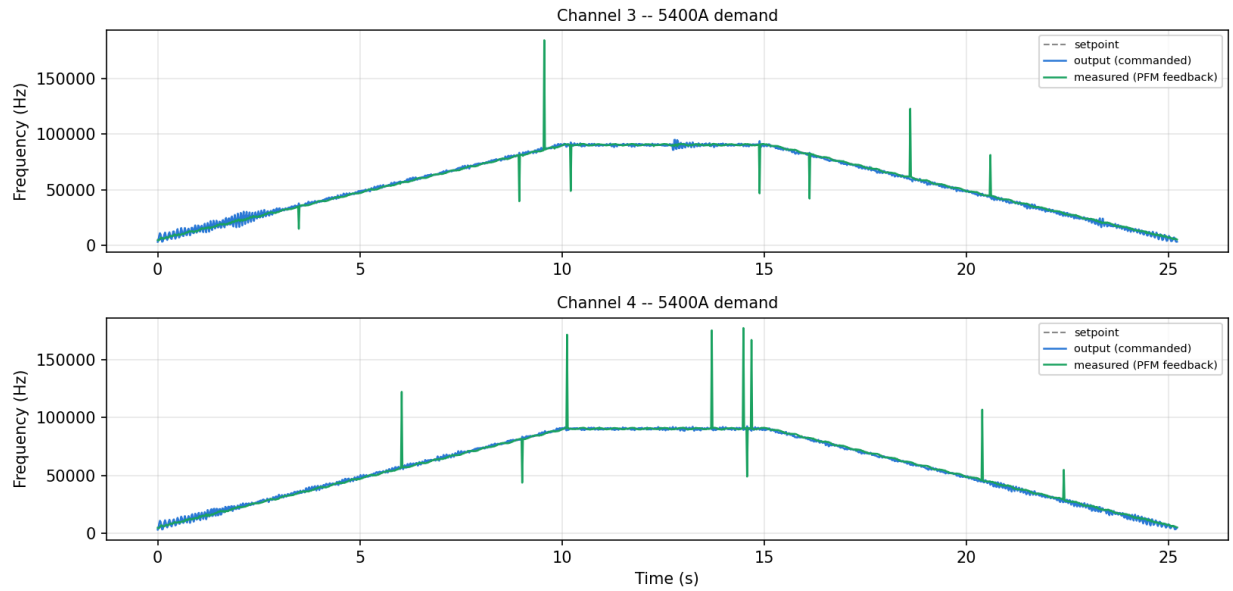
30 Shot 30: Channels 3+4, 5400 A demand

Gated channel(s) 3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 2 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch3=363Hz, ch4=409Hz.

Verdict: PASS

30.1 Per-channel response

Matrix: channels 3+4, 5400 A demand -- PASS (worst median error 409 Hz)



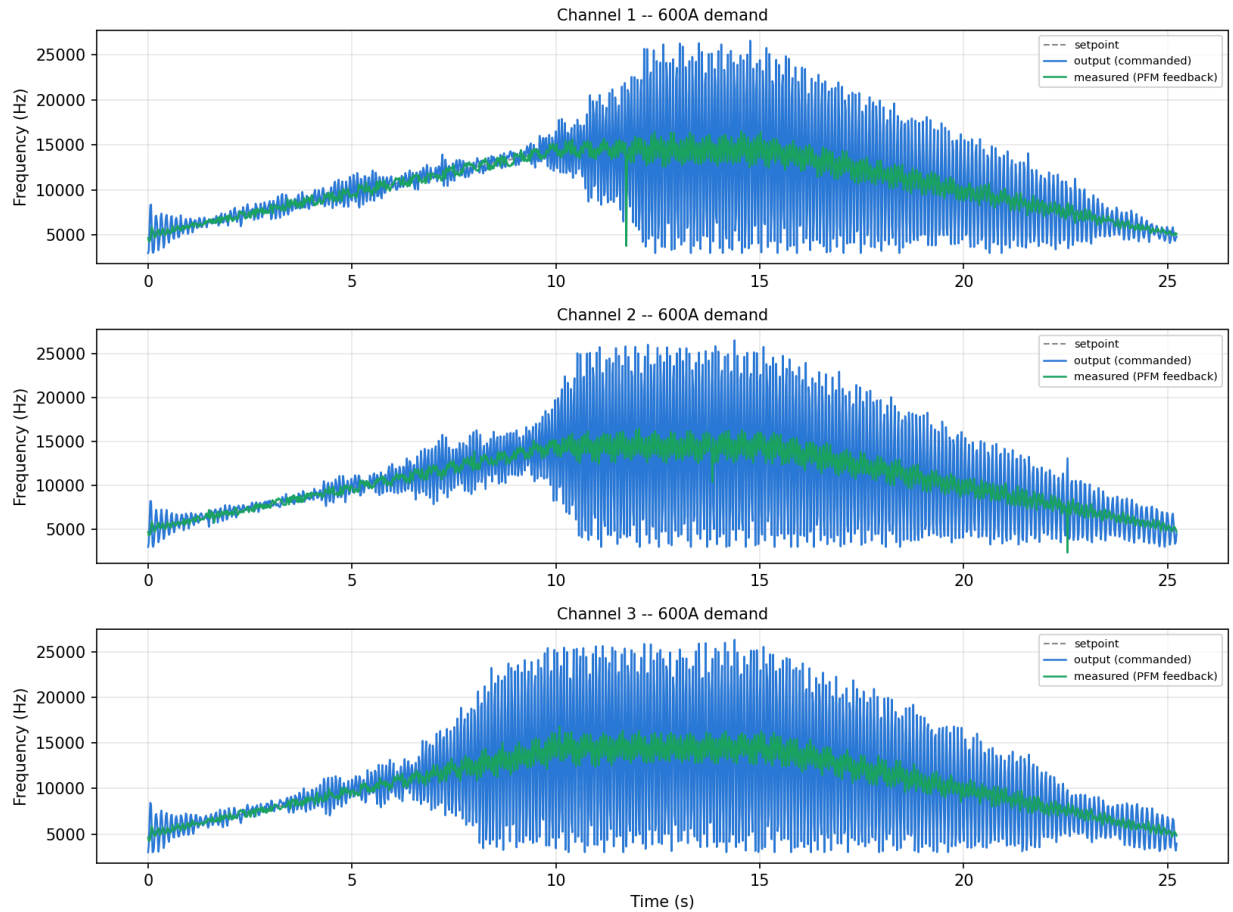
31 Shot 31: Channels 1+2+3, 600 A demand

Gated channel(s) 1+2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 909 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=739Hz, ch2=826Hz, ch3=909Hz.

Verdict: PASS

31.1 Per-channel response

Matrix: channels 1+2+3, 600 A demand -- PASS (worst median error 909 Hz)



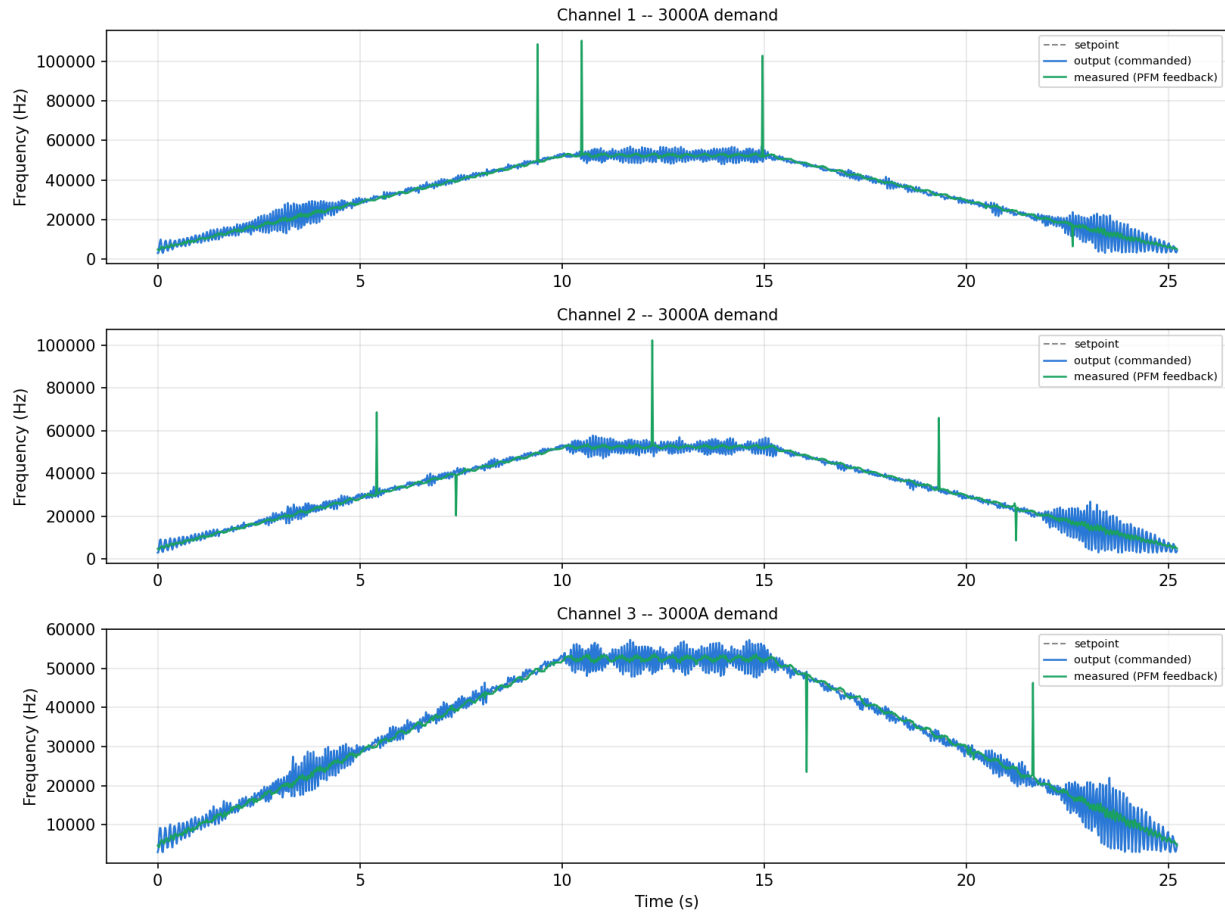
32 Shot 32: Channels 1+2+3, 3000 A demand

Gated channel(s) 1+2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 401 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=377Hz, ch2=360Hz, ch3=401Hz.

Verdict: PASS

32.1 Per-channel response

Matrix: channels 1+2+3, 3000 A demand -- PASS (worst median error 401 Hz)



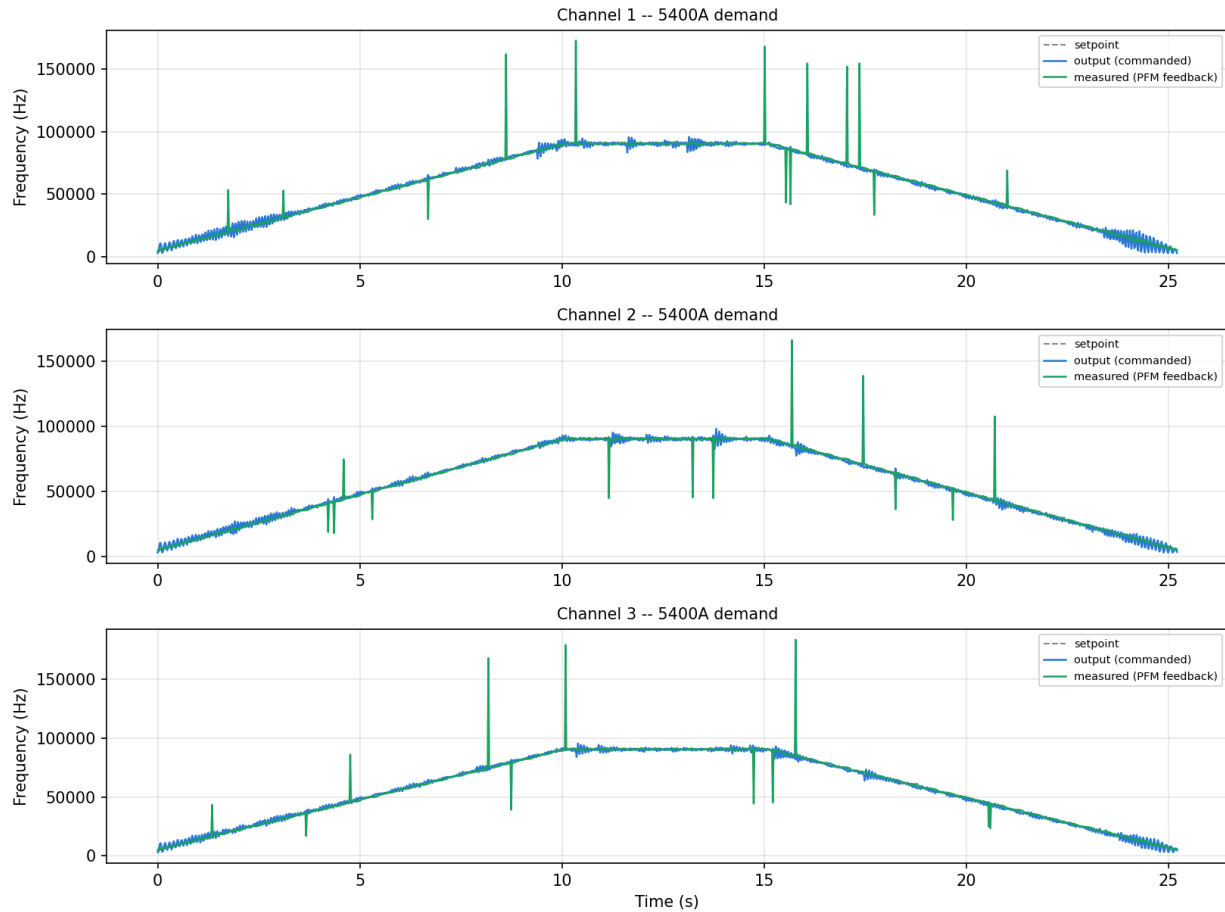
33 Shot 33: Channels 1+2+3, 5400 A demand

Gated channel(s) 1+2+3 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 363 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=363Hz, ch2=363Hz, ch3=363Hz.

Verdict: PASS

33.1 Per-channel response

Matrix: channels 1+2+3, 5400 A demand -- PASS (worst median error 363 Hz)



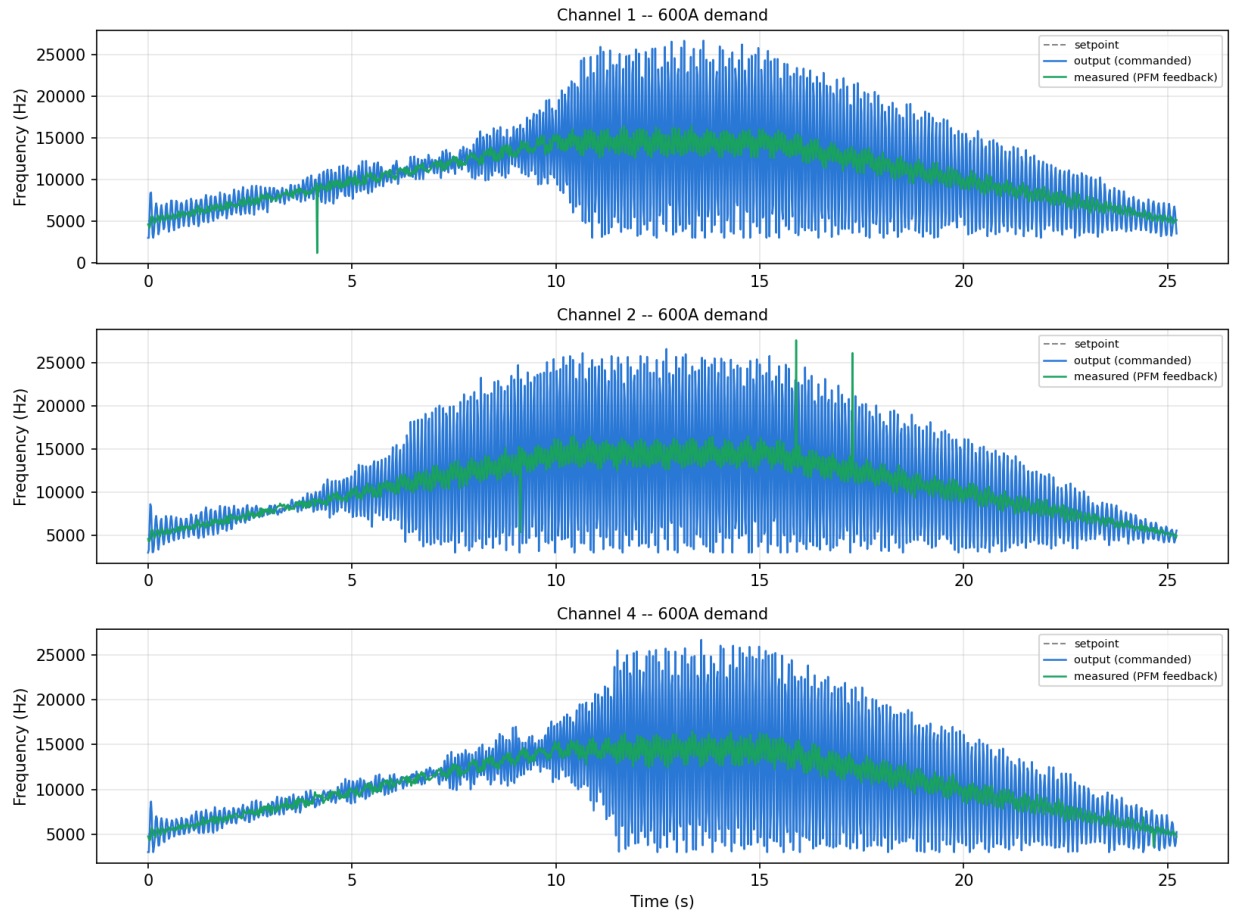
34 Shot 34: Channels 1+2+4, 600 A demand

Gated channel(s) 1+2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 906 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=887Hz, ch2=906Hz, ch4=738Hz.

Verdict: PASS

34.1 Per-channel response

Matrix: channels 1+2+4, 600 A demand -- PASS (worst median error 906 Hz)



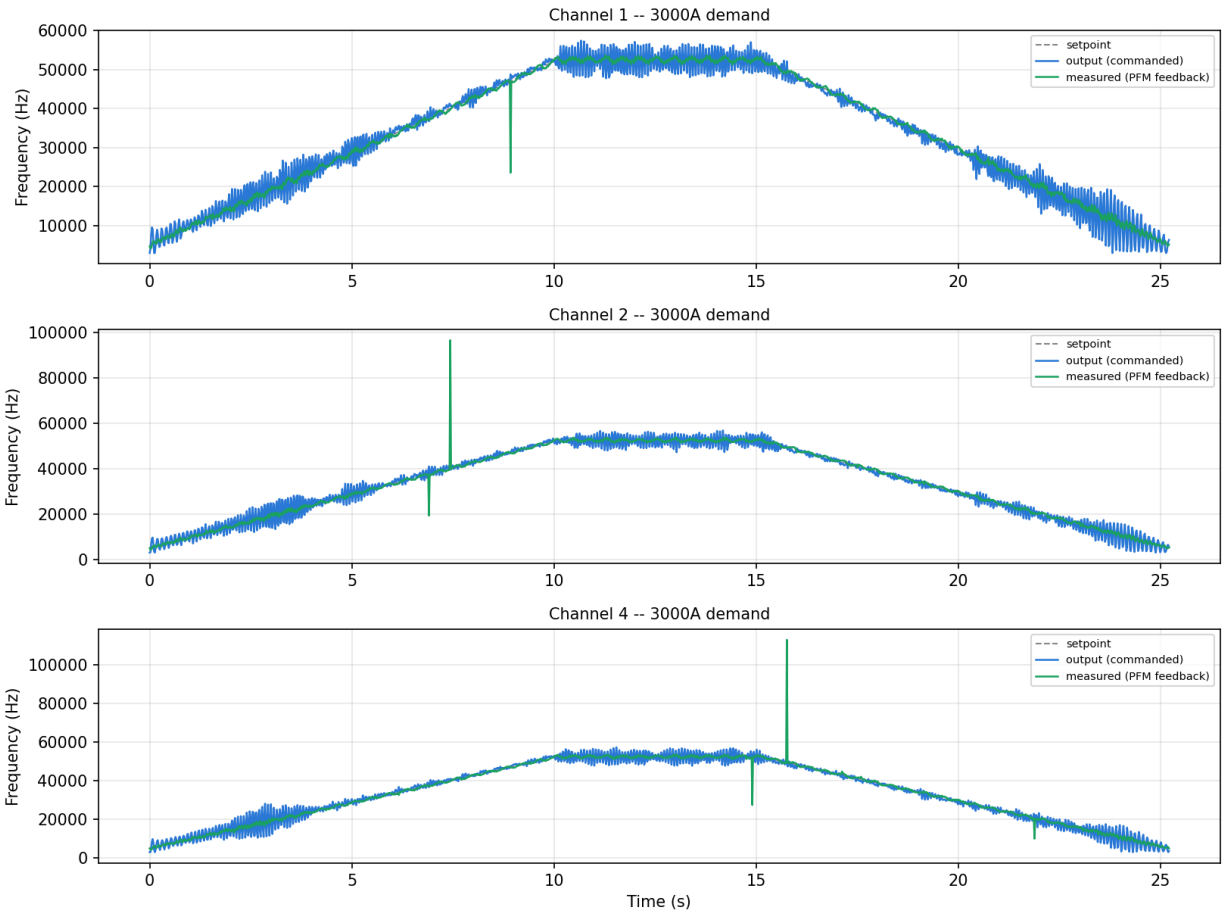
35 Shot 35: Channels 1+2+4, 3000 A demand

Gated channel(s) 1+2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 385 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=385Hz, ch2=377Hz, ch4=369Hz.

Verdict: PASS

35.1 Per-channel response

Matrix: channels 1+2+4, 3000 A demand -- PASS (worst median error 385 Hz)



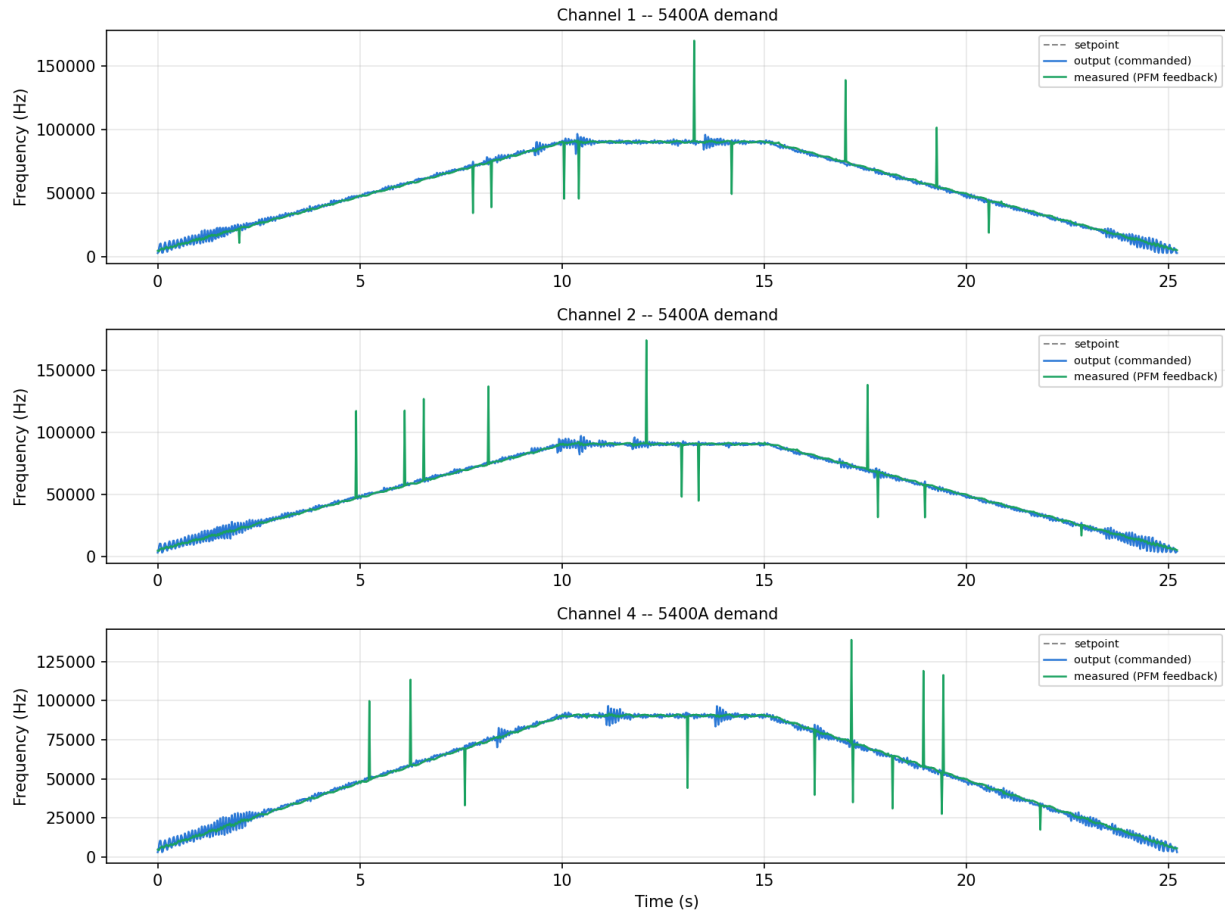
36 Shot 36: Channels 1+2+4, 5400 A demand

Gated channel(s) 1+2+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=409Hz, ch2=409Hz, ch4=363Hz.

Verdict: PASS

36.1 Per-channel response

Matrix: channels 1+2+4, 5400 A demand -- PASS (worst median error 409 Hz)



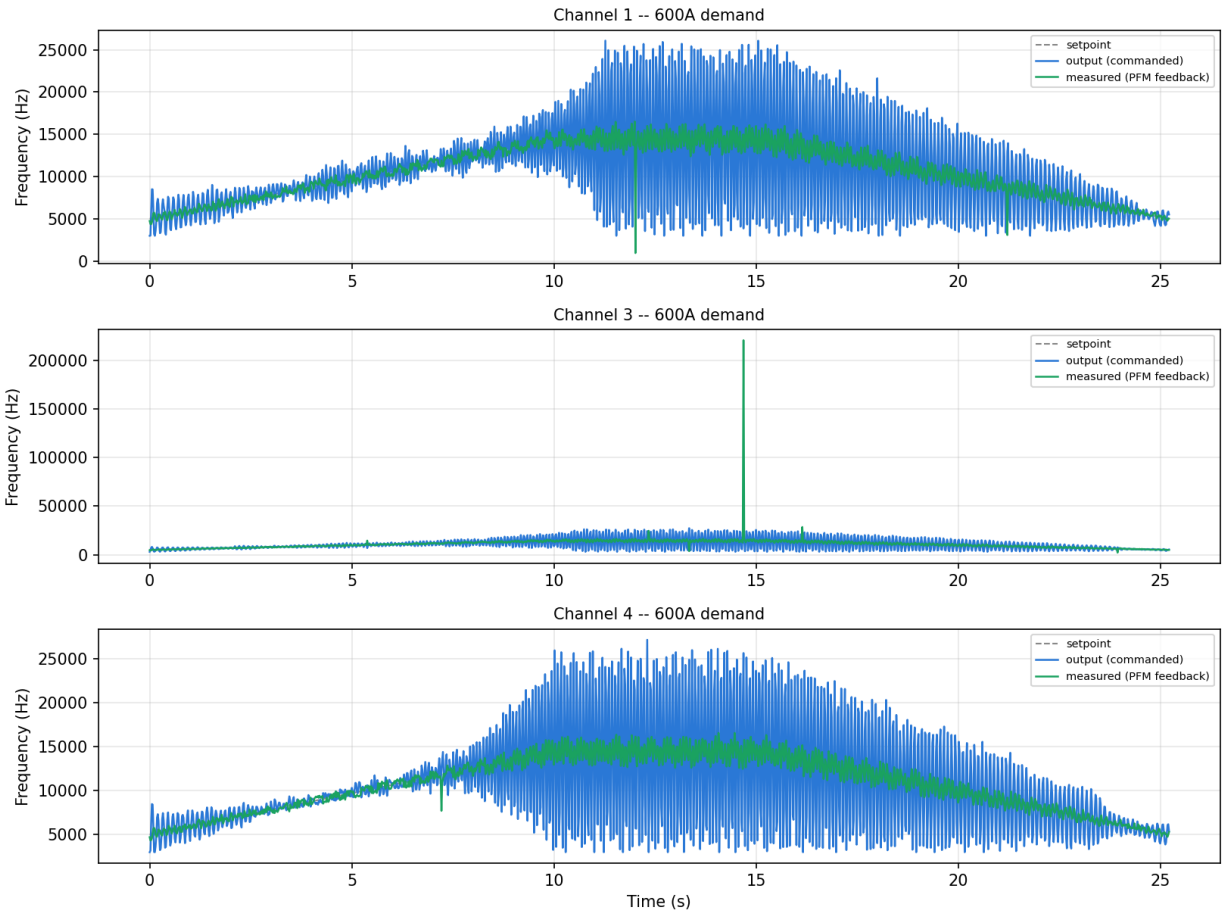
37 Shot 37: Channels 1+3+4, 600 A demand

Gated channel(s) 1+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 913 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=833Hz, ch3=913Hz, ch4=902Hz.

Verdict: PASS

37.1 Per-channel response

Matrix: channels 1+3+4, 600 A demand -- PASS (worst median error 913 Hz)



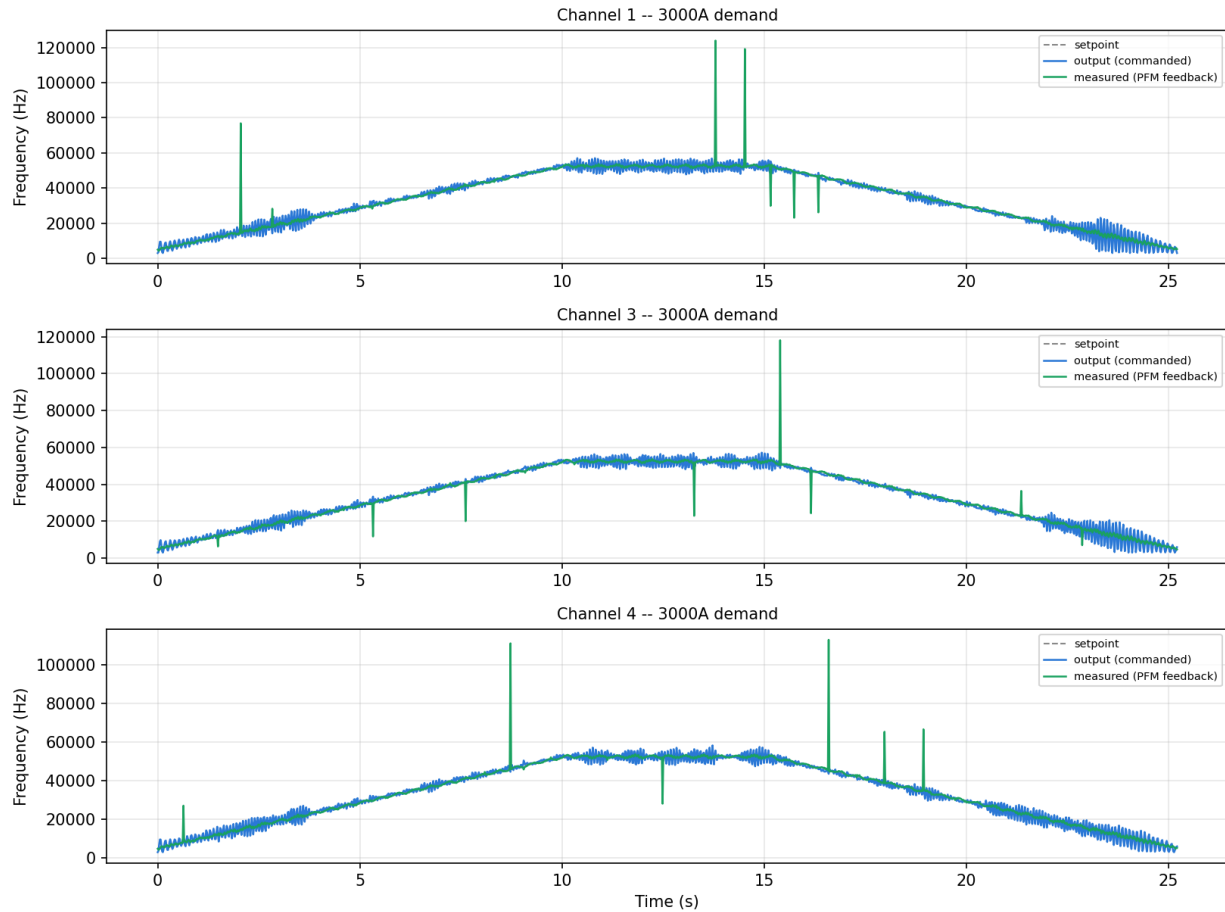
38 Shot 38: Channels 1+3+4, 3000 A demand

Gated channel(s) 1+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 377 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=369Hz, ch3=377Hz, ch4=360Hz.

Verdict: PASS

38.1 Per-channel response

Matrix: channels 1+3+4, 3000 A demand -- PASS (worst median error 377 Hz)



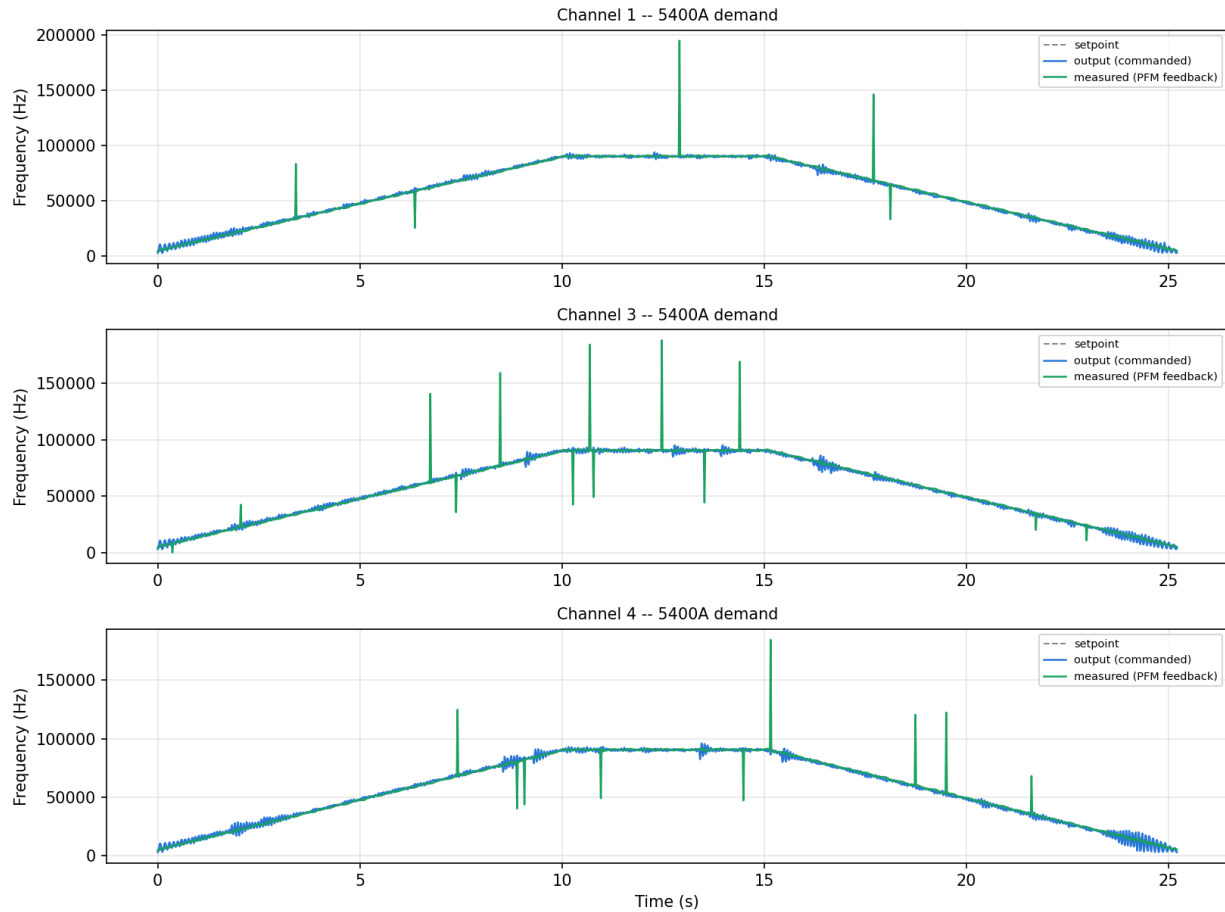
39 Shot 39: Channels 1+3+4, 5400 A demand

Gated channel(s) 1+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 363 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=363Hz, ch3=360Hz, ch4=363Hz.

Verdict: PASS

39.1 Per-channel response

Matrix: channels 1+3+4, 5400 A demand -- PASS (worst median error 363 Hz)



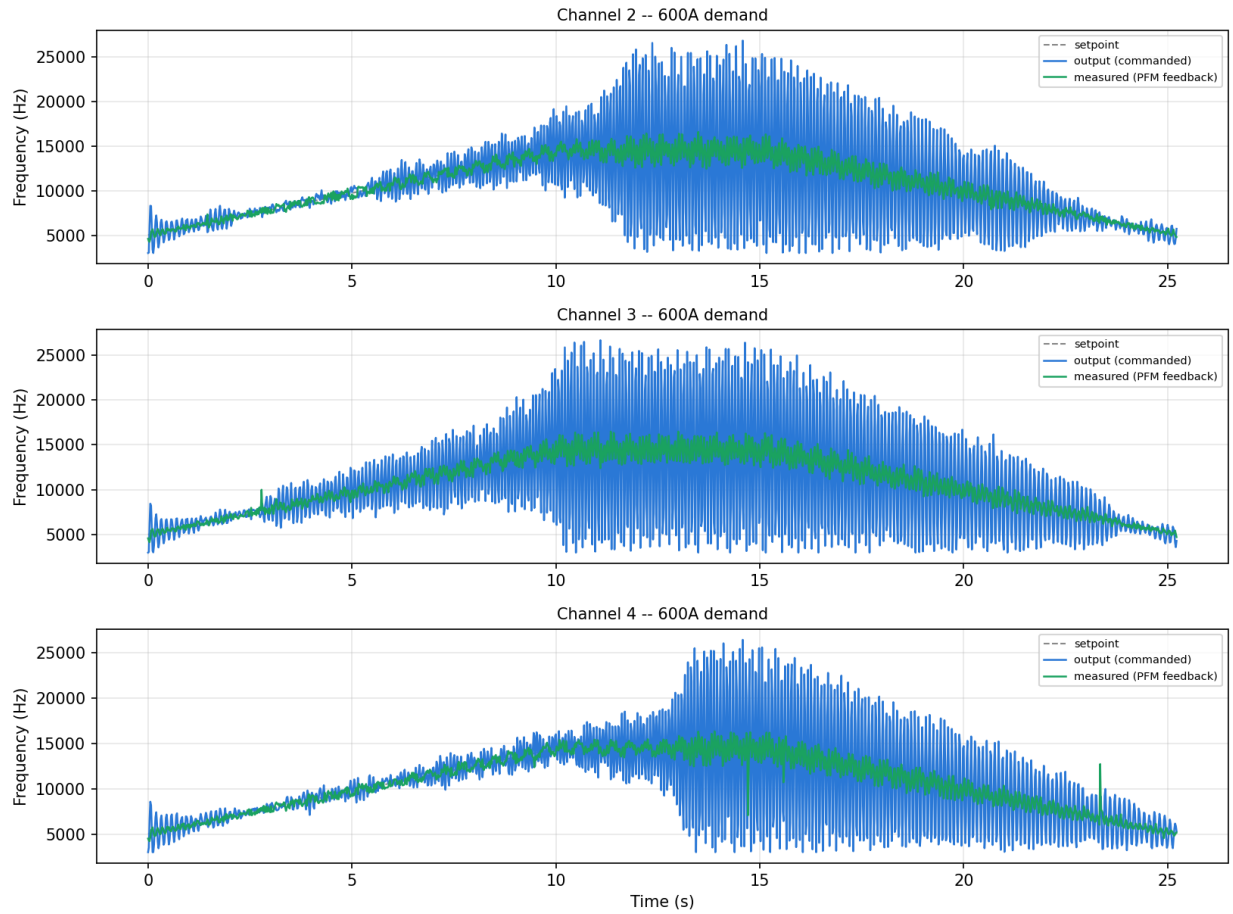
40 Shot 40: Channels 2+3+4, 600 A demand

Gated channel(s) 2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 851 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=732Hz, ch3=851Hz, ch4=518Hz.

Verdict: PASS

40.1 Per-channel response

Matrix: channels 2+3+4, 600 A demand -- PASS (worst median error 851 Hz)



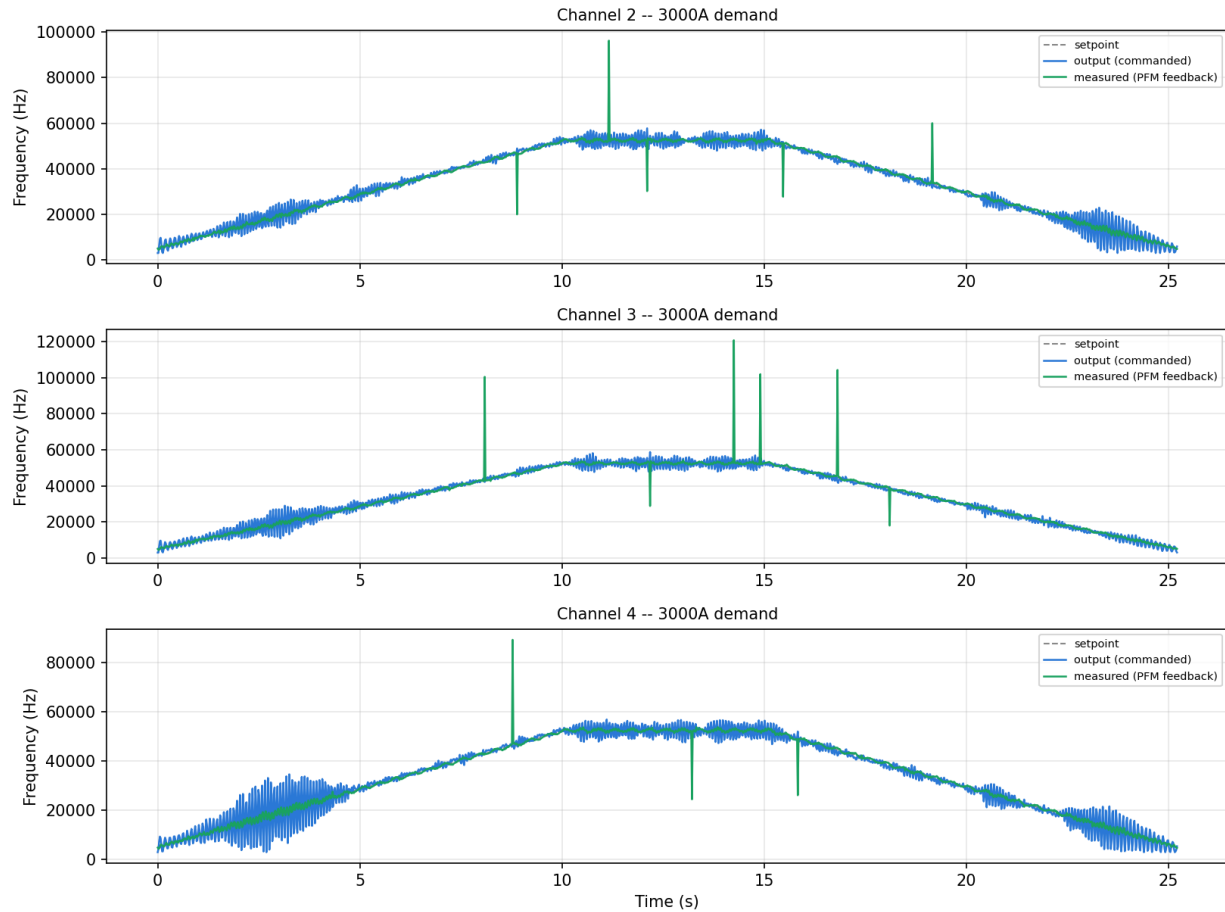
41 Shot 41: Channels 2+3+4, 3000 A demand

Gated channel(s) 2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 385 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=377Hz, ch3=385Hz, ch4=360Hz.

Verdict: PASS

41.1 Per-channel response

Matrix: channels 2+3+4, 3000 A demand -- PASS (worst median error 385 Hz)



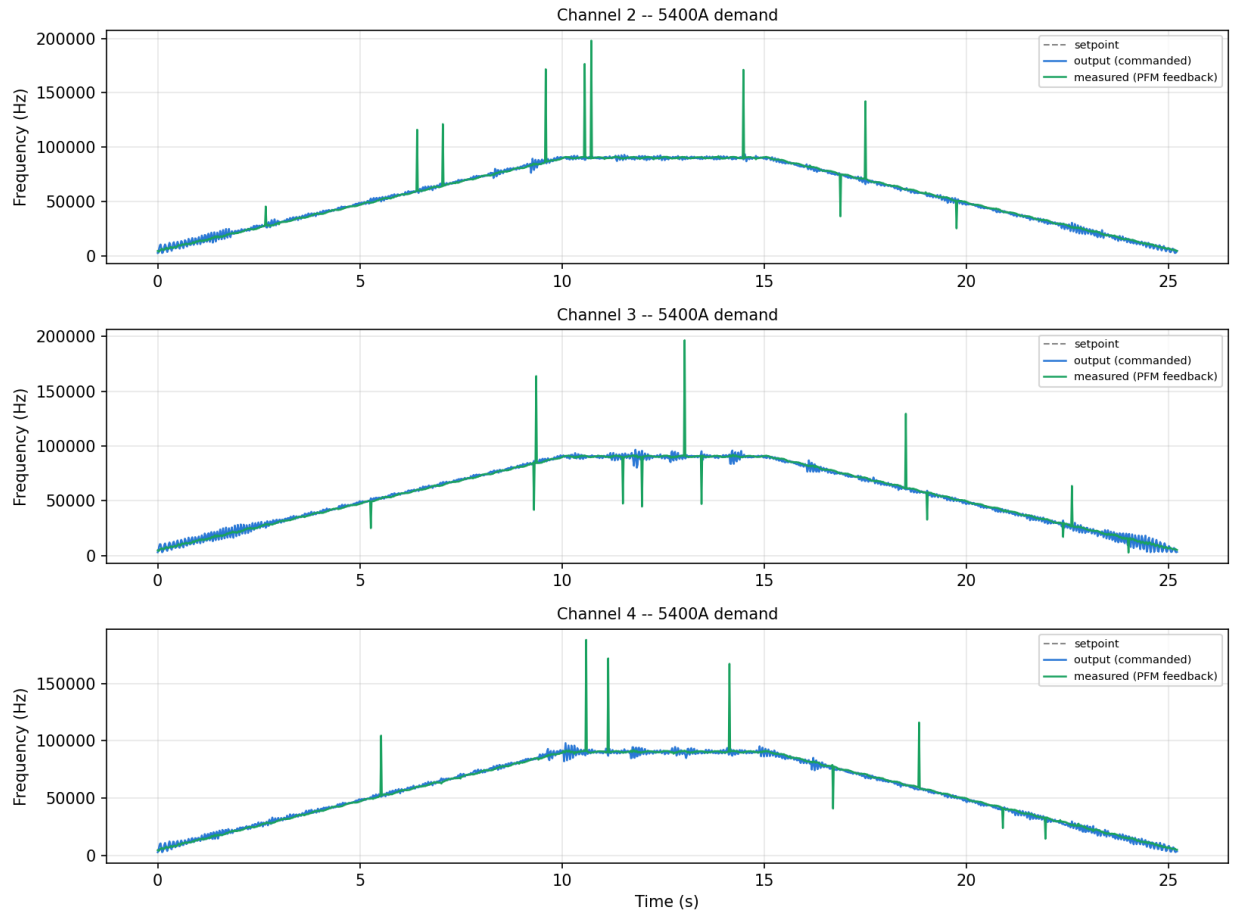
42 Shot 42: Channels 2+3+4, 5400 A demand

Gated channel(s) 2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 3 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 410 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch2=410Hz, ch3=409Hz, ch4=409Hz.

Verdict: PASS

42.1 Per-channel response

Matrix: channels 2+3+4, 5400 A demand -- PASS (worst median error 410 Hz)



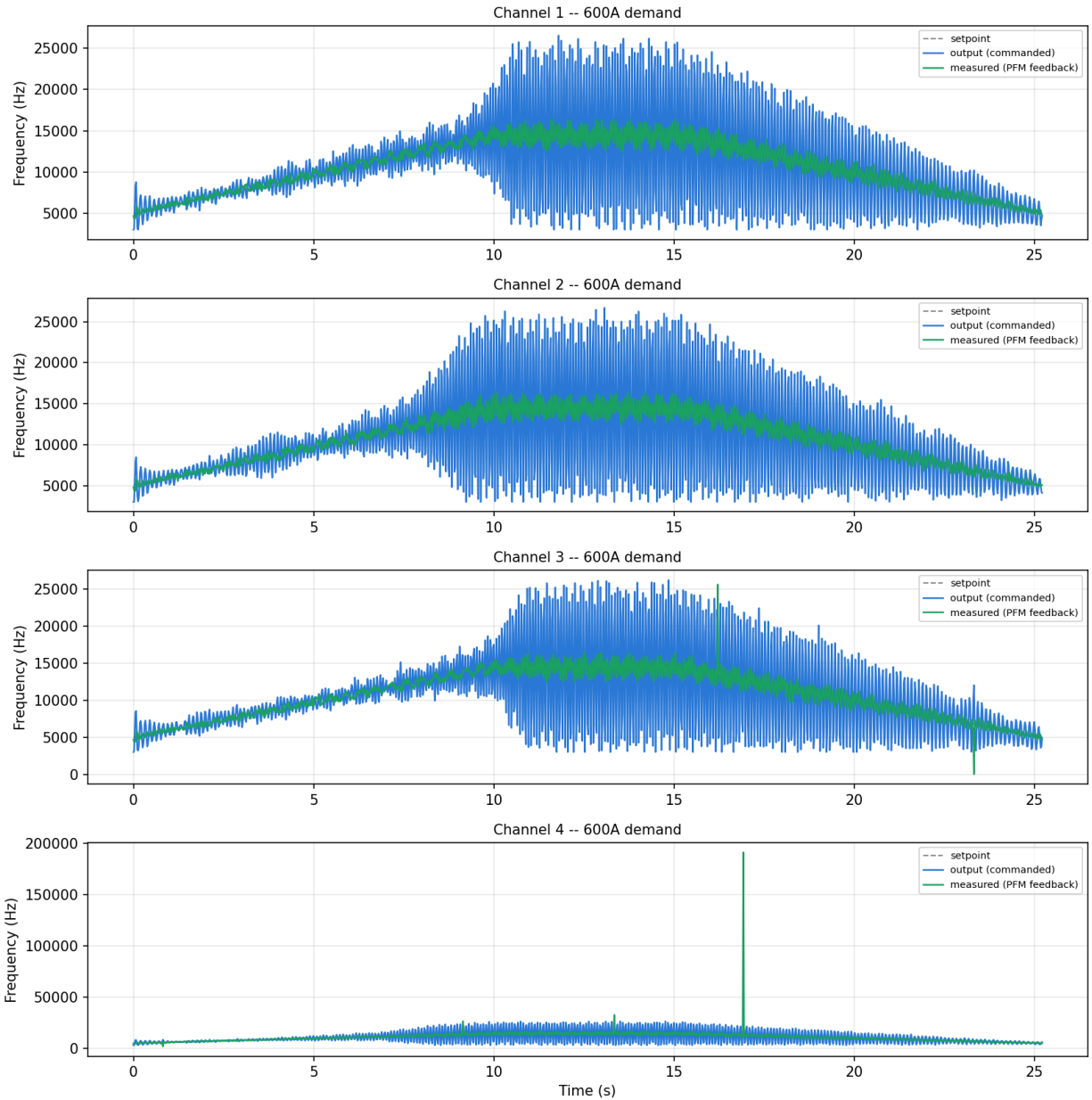
43 Shot 43: Channels 1+2+3+4, 600 A demand

Gated channel(s) 1+2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 600 A (14500 Hz) peak demand on every gated channel. Every one of the 4 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 890 Hz) – genuine closed-loop convergence to the commanded 14500 Hz setpoint (600 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=852Hz, ch2=851Hz, ch3=855Hz, ch4=890Hz.

Verdict: PASS

43.1 Per-channel response

Matrix: channels 1+2+3+4, 600 A demand -- PASS (worst median error 890 Hz)



44 Shot 44: Channels 1+2+3+4, 3000 A demand

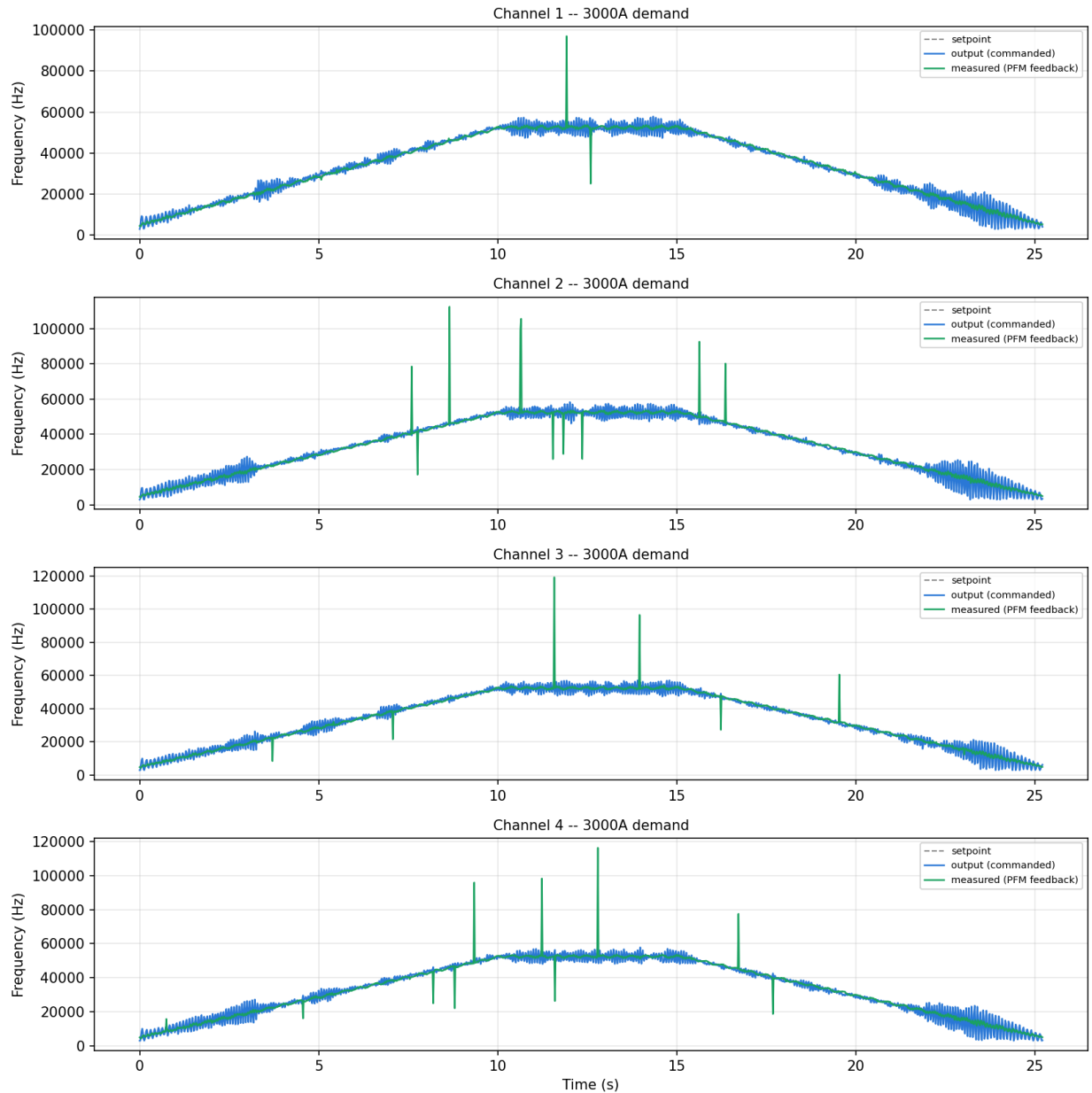
Gated channel(s) 1+2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 3000 A (52500 Hz) peak demand on every gated channel. Every one of the 4 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 401 Hz) – genuine closed-loop convergence to the commanded 52500 Hz setpoint (3000 A),

not just the controller's own commanded output following itself. flat-top median | measured-setpoint | per channel: ch1=401Hz, ch2=377Hz, ch3=401Hz, ch4=360Hz.

Verdict: PASS

44.1 Per-channel response

Matrix: channels 1+2+3+4, 3000 A demand -- PASS (worst median error 401 Hz)



45 Shot 45: Channels 1+2+3+4, 5400 A demand

Gated channel(s) 1+2+3+4 simultaneously, closed loop ($K_p=0.5$, $K_i=750$, $K_d=0$), commanding a real 10s/5s/10s ramp/flat-top/ramp-down profile to 5400 A (90500 Hz) peak demand on every gated channel. Every one of the 4 channel(s) in this combination converged, at flat-top, to within its own median $|measured - setpoint|$ error under the 1500 Hz pass threshold (worst channel: 409 Hz) – genuine closed-loop convergence to the commanded 90500 Hz setpoint (5400 A), not just the controller's own commanded output following itself. flat-top median $|measured - setpoint|$ per channel: ch1=409Hz, ch2=409Hz, ch3=363Hz, ch4=363Hz.

Verdict: PASS

45.1 Per-channel response

Matrix: channels 1+2+3+4, 5400 A demand -- PASS (worst median error 409 Hz)

